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How Do Monetary Authorities Buy Publicly Traded Financial Assets? A Cross-Country Database and an Empirical Study of Central Bank Asset Purchase Programs

Zhongxia Zhang and Yuhua Cai

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ABSTRACT: Despite the increasing use of asset purchase programs (APPs) by central banks over the past two decades, the economic literature has not empirically investigated how APPs are conducted from a cross-country perspective. Furthermore, a complete picture and critical details of APPs around the world remains lacking, particularly for small advanced economies and emerging markets. This paper fills this information gap by constructing a comprehensive global database on central bank asset purchase programs for the entire universe of advanced economies and emerging markets with monetary autonomy from 2008 to 2024. Our findings show that most central banks have conducted APPs, which have displayed heterogeneous characteristics, witnessed greater usage, and covered major asset classes beyond conventional government debt securities. The empirical analysis suggests that while triggers of APPs during 2008-2009 seem to be idiosyncratic across economies, disinflationary risks and government bond market volatilities activated the use of APPs in 2020. Furthermore, central banks that are constrained by the Zero Lower Bound and operate in countries with high public debt to GDP ratios tend to have longer buying durations of APPs. Finally, the pace of APPs depends on their previous pace as well as real GDP growth and CPI inflation and differs between buying and unwinding phases.

JEL Classification Numbers:	E44, E52, E58, F41, G15.
Keywords:	central bank asset purchase programs, unconventional monetary policy, quantitative easing, quantitative tightening, central bank balance sheet policies.

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September 26, 2026

Abstract

Despite the increasing use of asset purchase programs (APPs) by central banks over the past two decades, the economic literature has not empirically investigated how APPs are conducted from a cross-country perspective. Furthermore, a complete picture and critical details of APPs around the world remains lacking, particularly for small advanced economies and emerging markets. This paper fills this information gap by constructing a comprehensive

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global database on central bank asset purchase programs for the entire universe of advanced economies and emerging markets with monetary autonomy from 2008 to 2024. Our findings show that most central banks have conducted APPs, which have displayed heterogeneous characteristics, witnessed greater usage, and covered major asset classes beyond conventional government debt securities. The empirical analysis suggests that while triggers of APPs during 2008-2009 seem to be idiosyncratic across economies, disinflationary risks and government bond market volatilities activated the use of APPs in 2020. Furthermore, central banks that are constrained by the Zero Lower Bound and operate in countries with high public debt to GDP ratios tend to have longer buying durations of APPs. Finally, the pace of APPs depends on their previous pace as well as real GDP growth and CPI inflation and differs between buying and unwinding phases.

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Highlights

• This paper presents a comprehensive database of central bank asset purchase programs across advanced economies and emerging markets.

• Central bank asset purchase programs have expanded extensively, exhibited heterogeneous characteristics, and entered into major asset classes.

• The triggers, durations, and paces of central bank asset purchase programs can be explained by conventional macroeconomic and financial factors.

1 Introduction

Asset Purchase Programs (APPs) have increasingly become part of central banks' regular toolkits. The 2008 Global Financial Crisis severely shook financial systems and battered economic health in many countries. In response, many central banks aggressively cut policy rates to save the economy from taking a nosedive. As short-term interest rates approached the Zero Lower Bound (ZLB) and policy space narrowed, a small number of advanced economy central banks resorted to so-called Unconventional Monetary Policies (UMPs) such as large-scale asset purchase programs at that time. Through buying assets such as government bonds, central banks pushed down long-term interest rates to stimulate their economies. More than ten years later, the COVID-19 pandemic marked a sea change in the use of asset purchase programs as part of central banks' emergency policy packages. Major central banks that had previously conducted APPs reactivated them, and many other developed and developing nation central banks entered the uncharted territory of balance sheet expansions via asset purchases.

Despite the extensive use of asset purchase programs by many central banks in the last two decades, there is no empirical study on how central banks conduct APPs from a cross-country perspective. Prior studies on how APPs are carried out are descriptive and tend to focus on individual central banks or a limited set of central banks of systemic importance. For example, Joyce et al. (2011b) review the motivation for the Bank of England's asset purchases and describe the design and implementation of such programs. Neely and Fawley (2013) detail characteristics of the asset purchase programs of the Federal Reserve, Bank of England, European Central Bank, and Bank of Japan after the Global Financial Crisis. A recent study by Du, Forbes, and Luzzetti (2024) focuses on documenting central bank balance sheet reductions by seven large AE central banks in the post-COVID era and the associated macroeconomic and financial impacts. Without econometrically tested evidence, it is difficult to conclude how APPs are truly implemented in practice and whether actual implementation closely aligns with central banks' public statements. Not surprisingly, central bank announcements were the primary source for analyzing key issues such as triggers, durations, and paces of APPs in the past.

Moreover, there is a lack of a complete picture and critical details of APPs, especially for small advanced economies (AEs) and emerging markets (EMs). BIS (2019a) summarizes APP experiences after the Global Financial Crisis by seven central banks and BIS (2019b) discusses APPs by six central banks during 2008-2016. The only exceptions that include a broad range of central banks' APPs include BIS (2023) and Rebucci, Hartley, and Jiménez (2021). The

BIS (2023) paper provides qualitative descriptions of APPs by 19 central banks in response to the COVID-19 crisis and highlights their features and effectiveness. Rebucci, Hartley, and Jiménez (2021) put together an APP announcement dataset by 21 central banks in early 2020 and evaluate the announcement effects on financial markets. Nevertheless, none of the above studies provides exhaustive coverage of APPs in terms of country sample, program history in its entirety, and quantitative amounts. Regrettably, while there are extensive studies on the economic and financial effects of APPs, little is known about the full picture of APPs by AE and EM central banks in an internationally comparable manner.

Consequently, several fundamental questions about central bank asset purchase programs have remained open. These questions include but are not limited to: How many AE and EM central banks have conducted APPs? What are the objectives, types of operations, market segment of purchases, and accounting standards under each program? What types of asset classes did central banks buy under each program? What is the amount of asset purchases under each program? When do the buying and unwinding phases of APPs occur, and how long do APPs last?

In the absence of a comprehensive cross-country central bank asset purchase program database, researchers typically rely on suboptimal proxy measures of APPs for cross-country studies. One common approach is using central bank announcements instead of actual purchase amounts (Rebucci, Hartley, and Jiménez 2021; Krishnamurthy and Vissing-Jorgensen 2011; Bauer and Rudebusch 2014; Bauer and Neely 2014; Gagnon et al. 2011; Joyce et al. 2011a; Christensen and Rudebusch 2012; Altavilla et al. 2021; Glick and Leduc 2012; Smith and Valcarcel 2023; Weale and Wieladek 2016; Hesse, Hofmann, and Weber 2018; Neely 2015). However, the main problem with this event study approach is that most APPs are adjusted afterwards based on economic and market developments and actual purchase amounts do not stick to the original announcements. In some cases, central banks initially only announce the total envelope of asset purchases without mentioning the pace of asset purchases. An alternative approach is to approximate APPs according to the size of the central bank balance sheet (Dedola et al. 2021). But this proxy measure is often flawed and can be applied only in limited cases, because certain assets such as foreign exchange reserves and funding-for-lending schemes are not purchases of domestic marketable securities from the markets. Using such a proxy measure would overestimate the size of APPs and incorrectly mix APPs with other types of central bank operations. One less problematic approach is to take the time series of a central bank's claims on the general government as APPs (Beirne and Sugandi 2023). Nevertheless, it is an imperfect solution, especially when central banks have bought other asset classes beyond government debt securities in their APP portfolios. The series of central

bank's claims on the general government does not capture frequently bought assets such as asset-backed securities and corporate bonds.

This paper offers a seminal contribution to the economics literature by constructing a global database of central bank asset purchase programs for the entire universe of advanced economies and emerging markets with monetary autonomy over the period of 2008–2024. We define APPs as central banks' purchases of domestic marketable securities from primary or secondary markets, excluding short-term borrowing or lending operations. We have devoted meticulous efforts to collecting information from each of the central banks' websites and closing remaining data gaps using mirror data sources. The compiled database not only comprises qualitative information on asset purchase programs' attributes, but also quantitative panel data series at monthly frequency.

This unique database documents each asset purchase program's characteristics, asset classes, net purchase amounts, and timelines. Our statistics suggest that most central banks in advanced economies and emerging markets (35 out of 48 central banks with monetary autonomy in our sample) have conducted some form of asset purchase programs during 2008-2024. While it was mostly AE central banks that conducted APPs before 2020, many EM central banks followed suit after the COVID-19 pandemic. Moreover, there are substantial variations in APP characteristics, including program objectives, types of operations, market of purchases, accounting standards, and data disclosures. Interestingly, many central banks have extended their APP portfolios beyond central government bonds. Additional instruments include state and municipal government securities, asset-backed securities, financial and nonfinancial corporate bonds, money market instruments, equities and equity Exchange-Traded Funds (ETFs), Real Estate Investment Trusts (REITs), and even central banks' own securities.

Another novel contribution of this research is the empirical analysis of the determinants of asset purchase programs' triggers, durations, and amounts. Utilizing this novel database, we conduct cross-country regressions to explore under what regimes and conditions central banks activate APPs, what initial factors determine the length of buying phases in APPs, and which economic and financial variables determine the pace of APPs during portfolio accumulation and unwinding stages. Our findings indicate that no common factor can explain triggers of APPs during 2008-2009, while disinflationary risks and government bond market volatilities activated the use of APPs during 2020. Furthermore, central banks that are under the Zero Lower Bound and in countries with high public debt to GDP ratios tend to have longer buying durations of APPs. Lastly, the pace of APPs depends on whether the program is in the buying phase or the unwinding phase. The pace of APPs during buying phases

depends on its pace in the previous period and real GDP growth, while the pace of APPs during unwinding phases depends on its pace in the previous period and CPI inflation.

2 Literature Review

Several prior cross-country studies examine the scope and effects of central bank asset purchase programs. BIS (2023) documents central bank asset purchases during the COVID-19 crisis. The report provides details on asset purchase programs by 19 major economies and discusses the features, effectiveness, and side effects of such programs. BIS (2019a) is a cross-country analysis of unconventional monetary policy tools and a part of this report reviews the experiences of asset purchase programs after the Global Financial Crisis by 7 central banks. BIS (2019b) investigates monetary policies and central bank balance sheet expansions including asset purchase programs. The report notes that 6 out of 23 central banks in a survey have launched asset purchase programs during 2008–2016. Du, Forbes, and Luzzetti (2024) analyze the cross-country experience in seven advanced economies and assess the macroeconomic and financial impact of shrinking central bank balance sheets due to quantitative tightening. Neely and Fawley (2013) describe the motivations for and characteristics of the asset purchase programs of the Federal Reserve, Bank of England, European Central Bank, and Bank of Japan during 2008-2012. Rebucci, Hartley, and Jiménez (2021) compile a dataset of 30 quantitative easing announcements made by 21 central banks in early 2020 and apply an event study approach to assess the impact of such announcements on government bond yields and exchange rates. Using data on central bank claims to the government as a proxy variable for asset purchase programs, Beirne and Sugandi (2023) explore the impact of these programs by 14 emerging market central banks during the COVID-19 crisis and find that they have persistent financial market effects. Eckhold et al. (2024) survey intervention measures including asset purchases by central banks in developing countries during the COVID-19 pandemic and find these measures improved market functioning and generated positive spillover effects. Cantú et al. (2021) present a dataset covering central bank announcements in 39 economies of five types of policy measures adopted in response to the COVID-19 pandemic, including asset purchase programs with qualitative details. Arena et al. (2021) provide an overview of asset purchase programs in six emerging European economies and discuss their effectiveness, constraints, and exit strategies.

The literature has suggested several transmission mechanisms for central bank asset purchase programs. An early work by Krishnamurthy and Vissing-Jorgensen (2011) identified various channels through which APPs may affect interest rates, including the signaling

channel, the duration risk channel, the liquidity channel, the safety channel, the prepayment risk premium channel, the default risk channel, and the inflation channel. D'Amico et al. (2012) adopt a framework to incorporate several transmission channels of the Federal Reserve's large-scale asset purchase programs and show that both the scarcity and duration channels are important. Koijen et al. (2017) and Koijen et al. (2021) use security-level holding data in the Euro Area to provide empirical evidence on the portfolio rebalancing channel. Bauer and Rudebusch (2014) find that the Federal Reserve's bond purchases have signaling effects that reduce expected future short-term interest rates. Di Maggio, Kermani, and Palmer (2020) reveal a refinancing channel by estimating household refinancing activities during the Federal Reserve's large-scale asset purchases. Bauer and Neely (2014) apply dynamic term structure models to distinguish signaling and portfolio balance channels that affect international bond yields.

Previous research has found significant effects of central bank asset purchase programs on domestic financial markets and exchange rates. Gagnon et al. (2011) find that the Federal Reserve's large-scale asset purchases led to economically meaningful and long-lasting reductions in long-term interest rates on U.S. securities. In the case of the United Kingdom, Joyce et al. (2011a) find the Bank of England's quantitative easing policy has depressed medium to long-term Gilt yields. Christensen and Rudebusch (2012)'s work on asset purchases by the Federal Reserve and the Bank of England also shows lower yields in the US and the UK. Similar reduction effects are found in European sovereign and corporate bond yields by the European Central Bank's asset purchases (Todorov 2020; Altavilla et al 2021). Evidence also points to depreciation of the U.S. dollar, the euro, and the British pound after announcements of asset purchases by the Federal Reserve, the European Central Bank, and the Bank of England (Glick and Leduc 2012; Dedola et al. 2021). Lastly, Smith and Valcarcel (2023)'s analysis indicates that unwinding past Fed asset purchases tightens financial conditions.

Researchers have also studied the effects of central bank asset purchase programs on the domestic economy. While acknowledging uncertainties in the magnitude of the impact, Joyce et al. (2011b) and Kapetanios et al. (2012) conclude that quantitative easing has boosted real GDP and inflation in the UK. In addition, Weale and Wieladek (2016) demonstrate that announcements of large-scale government bond purchases have a statistically positive impact on real GDP and inflation in the US and the UK. Rodnyansky and Darmouni (2017) show that U.S. banks with a relatively large exposure to mortgage-backed securities aggressively expanded their lending when the Fed targeted this type of security. Results by Hesse, Hofmann, and Weber (2018) suggest that the macroeconomic effects of early asset purchase programs were significantly positive, while the effects of subsequent programs were weaker.

Another strand of the literature focuses on the international spillover effects of central bank asset purchase programs. For instance, the Federal Reserve's announcements of asset purchases in 2008-2009 substantially reduced long-term bond yields in other developed countries, as shown in Neely (2015). Fratzscher, Lo Duca, and Straub (2016)'s work suggests that the Fed's quantitative easing increased procyclicality of portfolio flows outside the US, particularly into emerging market equities. Furthermore, Bhattarai, Chatterjee, and Park (2021)'s work shows that emerging markets experienced currency appreciations, bond yield reductions, and stock market booms due to U.S. quantitative easing. The study by Chari, Stedman, and Lundblad (2021) indicates that U.S. unconventional monetary policy had asymmetric effects on emerging market capital flows and asset prices, with active capital reallocations being more important in the policy unwinding period.

3 A Central Bank Asset Purchase Program Database

3.1 Data Compilation Methodology

In this paper, we define asset purchase programs as central banks' purchases of domestic marketable securities from primary or secondary markets, excluding short-term borrowing or lending operations.[1] This definition allows scholars to understand how central banks directly enter publicly traded financial markets through unconventional monetary policies. For instance, many countries' sovereign debt investor bases have dramatically changed, and central banks have become notable holders of sovereign debt as a result of APPs in the last two decades.

Under the above definition, several types of central bank operations are disqualified as asset purchase programs. The definition excludes conventional monetary policy operations and other unconventional monetary policy operations. Funding-for-lending schemes, another type of UMPs that provides low-cost central bank financing to commercial banks to boost their credit to the real economy, does not fit into our APP definition as the operation does not involve the exchange of marketable securities. Central bank repos, reverse repos, discount windows, and term deposit facilities do not count because they are borrowing or lending activities rather than asset purchases or sales. Foreign exchange interventions, a conventional monetary policy tool commonly used by developing country central banks to affect the level or volatility of exchange rates, are excluded because such interventions typically require

[1] Open market operations that aim to keep the policy rate around the target or within the corridor set by a central bank are not regarded as asset purchase programs.

purchases or sales of foreign securities. Furthermore, bank recapitalization by a central bank through an equity injection, accompanied by the acquisition of bank bonds in return, is not considered an APP because the policy intention is to restore bank capital adequacy and the acquired securities are often not marketable. Similarly, central bank loans, receivables, or overdrafts are not included in APPs since no exchange of marketable securities is involved. Finally, central banks in certain countries act as Finance Ministries' fiscal agents to perform Treasury buyback programs and other Liability Management Operations (LMOs) on behalf of governments. These LMOs are not APPs given that these transactions aim to achieve the government's public debt management strategy and they are recorded under the government (the ultimate holder)'s balance sheet.

We perform a screening process that covers the entire universe of advanced economy and emerging market central banks. First, we restrict the sample to central banks that have monetary autonomy. Our definition of monetary autonomy follows the International Monetary Fund's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER), which classifies a country's monetary policy framework under one of four categories: exchange rate anchor, inflation targeting, monetary aggregate targeting, and other. If a central bank maintains an exchange rate peg as its nominal anchor, it is considered to lack monetary autonomy because a fixed exchange rate regime requires the domestic central bank to follow monetary policy in the anchor currency country to maintain the peg. The rationale for excluding those central banks without monetary autonomy[2] is the assumption that they do not conduct asset purchase programs for monetary policy purposes.[3] Second, we consider all Eurozone countries[4] as one economy in our database because the Governing Council has the monetary policy decision-making power for the entire Euro System. The above criteria leave a sample of 48 AE and EM central banks with monetary policy autonomy.

We examine each of the 48 central banks' websites and other official sources to check whether these monetary authorities have conducted APPs during 2008-2024. We comb through central bank sources including dedicated webpages about their asset purchase programs, central bank balance sheets and financial statements, monetary policy statements and reports, and central bank annual reports. When additional information is needed, we also rely on mirror data sources such as debt-related publications on the Ministry of Finance's website.

We follow a decision tree method to construct aggregate APP time series at monthly

[2]These exchange rate anchor economies include Azerbaijan, Bulgaria, Denmark, Hong Kong SAR, Iran, Morocco, Panama, Qatar, Saudi Arabia, and United Arab Emirates.

[3]Nevertheless, it is possible that central banks under exchange rate pegs may intervene in the markets for market functioning purpose.

[4]The sample includes Croatia as a separate economy before its euro adoption in 2023.

frequency by asset class using public data sources.[5] The process flow in Figure 1 is applied to each economy from top to bottom. For central banks that have conducted APPs, each is assigned to a mutually exclusive group. First, we examine whether they publish asset purchase data. There are 6 central banks that set up dedicated webpages or balance sheet sections to disclose APPs. A more common case is that central banks publish securities holdings data in their balance sheets. 24 central banks fall into this group. Another case is that central banks neither post APP data nor securities holdings data. For 3 central banks in this group, we rely on their balance sheet items such as "central bank claims on the government sector" and "central bank claims on the corporate sector" to approximate APPs. For cases that rely on central bank balance sheet information due to a lack of APP data disclosure, we subtract stock values of securities holdings or claims just before APP initiation periods to derive our proxies of APPs. These adjustments are intended to exclude conventional monetary policies such as repo operations from the APP time series.[6] Lastly, there are 2 central banks that have only bought government debt securities in their APPs but do not publish such information. Mirror data sources are used to fill the APP time series in this case. The list of central banks in each group used in the data collection process is provided in the appendix. In the supplementary note, we give a detailed description of the data sources and compilation methods for each central bank's APP time series.

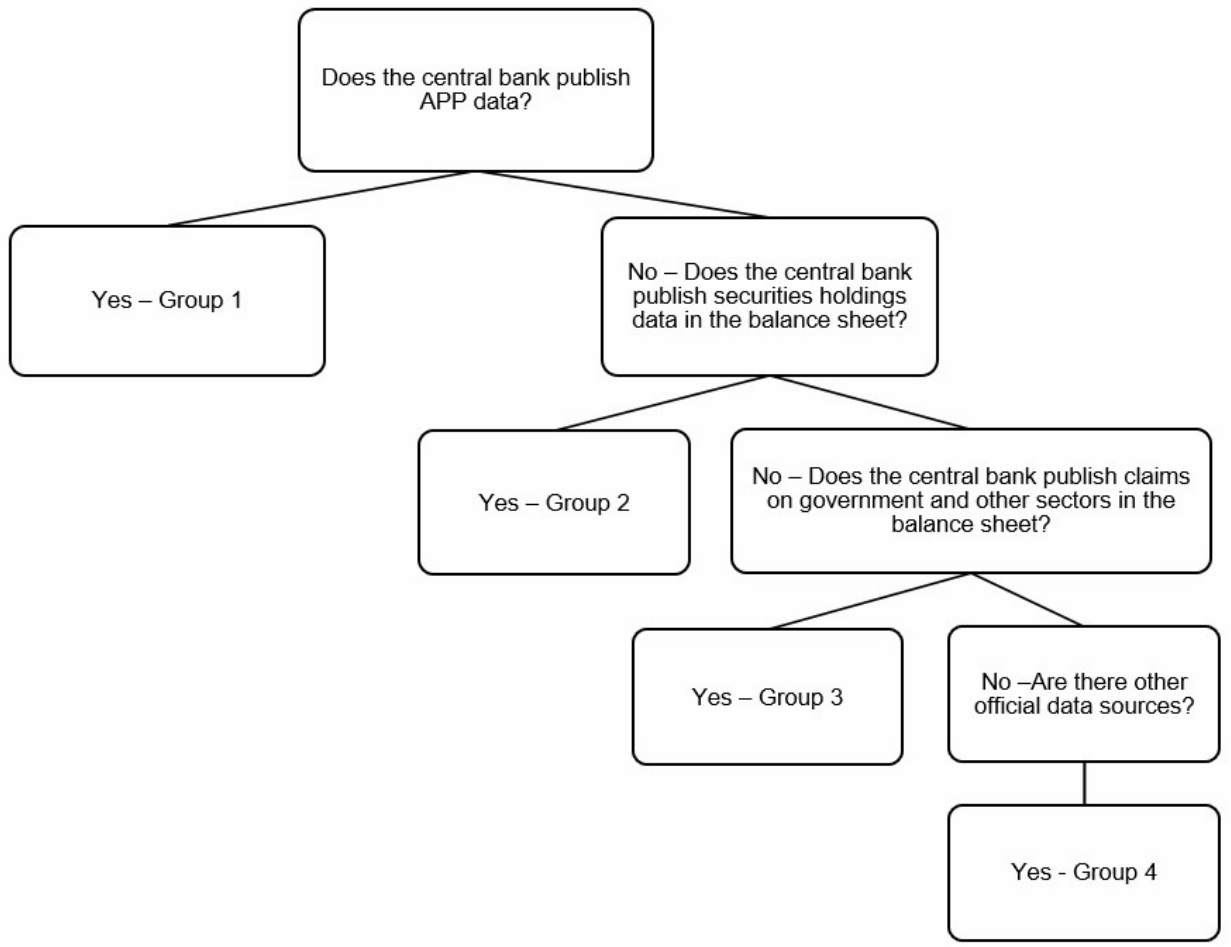
We date APP timelines based on central bank announcements and the constructed APP time series. Our coding rules give precedence to central bank announcements whenever such public statements are available.[7] In the absence of central bank announcements on APPs, we apply a local maximum identification algorithm (i.e., identifying the maximum level of APP holdings within each economic cycle, as defined by real GDP expansions and recessions) to distinguish APP cycles. Under this approach, net buying phases precede the peak in securities holdings, while unwinding phases follow shortly thereafter. Periods during which APP securities holdings remain unchanged are classified as neither buying nor unwinding phases. We then construct a categorical variable to indicate buying phases, unwinding phases, and neither buying nor unwinding phases.

[5]Our APP time series are recorded on a net basis, measuring the net amount of asset purchases rather than gross purchases, gross redemptions, or gross sales. This distinction is important because a central bank may simultaneously purchase and sell securities, or acquire new securities while only partially reinvesting the proceeds from maturing securities.

[6]The APP time series constructed for Groups 2 and 3 central banks reflect the best possible efforts. Admittedly, they are approximations instead of perfect measures of APPs.

[7]Note that central banks' announcements may not perfectly correspond to the magnitude or direction of actual asset purchases.

Figure 1: APP Data Collection Decision Tree



Note: The figure plots the data collection method to construct aggregate APP time series. We follow a decision tree under a series of guided questions to identify public data sources. The process flows from top to bottom and each economy with available data ends in a mutually exclusive group.

3.2 Stylized Facts

In our data sample, most central banks have conducted some form of asset purchase programs during 2008-2024. In total, 35 out of 48 central banks have introduced APPs. A typical APP occurs around or after crisis periods such as the 2008 Global Financial Crisis and the COVID-19 pandemic. However, there are also special country cases: Argentina's central bank purchased government bonds to provide monetary financing to the government. Ukraine's central bank bought government war bonds from the primary market during its war with Russia in 2022. Part of the APP by the Central Bank of Sri Lanka coincided with a sovereign debt default in the spring of 2022.

Central bank asset purchase programs have shown heterogeneous characteristics in their objectives, asset sales, types of operations, market of purchases, accounting standards, and data disclosures (Table 1). Central banks do not often disclose their APP data in detail, which makes our database a valuable contribution to economic literature. Central banks usually cite monetary policy, market functioning, or both as objectives for their APPs. APPs are considered useful monetary policy tools to put downward pressure on long-term interest rates when the policy rate is low or at the Zero Lower Bound. In addition, central banks rely on APPs to ensure that market functioning returns to normal when disorderly market conditions emerge. Most central banks unwind their portfolio holdings sometime after initiation of their APPs. In terms of portfolio unwinding strategy, some central banks both actively and passively cut their holdings while other central banks just passively reduce their balance sheet sizes by partially reinvesting maturing securities or completely ending the rollover of maturing securities.[8] Besides changing the sizes of APP portfolio holdings, certain central banks have deployed operation twists by simultaneously selling short-term Treasury securities and buying long-term Treasury securities to influence sovereign yield curves. Moreover, although the majority of central banks implement their APPs through the secondary market, a few central banks have bought assets from both the primary and secondary markets as secondary markets in their economies are typically shallow with inadequate liquidity to absorb sizable transactions. Finally, the accounting standards adopted by central banks to record their APPs vary significantly, ranging from face value to amortized cost and to market value accounting treatments.[9]

While government debt securities are the most widely bought financial assets, central

[8] An active unwind refers to the sale of securities before maturity, whereas a passive unwind occurs when the proceeds from maturing securities are not fully reinvested.

[9] Differences in accounting treatments make our APP series not fully comparable across countries, since valuation changes are properly reflected under fair value accounting but not under face value or amortized cost accounting.

Table 1: Summary of APP Characteristics by Country

Country	Time Series Data Disclosure	Stated Objective(s)	Portfolio Unwind	Unwind Method	Operation Twist	Market of Purchase	Accounting Standard
United States	No	Monetary policy; Market functioning	Yes	Passive	Yes	Secondary market	Face value
United Kingdom	Yes	Monetary policy; Market functioning	Yes	Active and passive	No	Secondary market	Face value
Euro Area	Yes	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Sweden	No	Monetary policy; Market functioning	Yes	Active and passive	No	Secondary market	Face value
Canada	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Switzerland	No	Market functioning	No	N.A.	No	Secondary market	Issue price 1/
Japan	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost 2/
Türkiye	No	Market functioning	Yes	Active and passive	No	Secondary market	Amortized cost
Australia	Yes	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Face value
New Zealand	Yes	Monetary policy; Market functioning	Yes	Active and passive	No	Secondary market	Fair value
South Africa	No	Market functioning	Yes	Passive	No	Secondary market	Fair value
Chile	Yes	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Colombia	No	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Mexico	Yes	Market functioning	No	N.A.	Yes	Secondary market	Market value
Israel	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
India	No	Monetary policy; Market functioning	Yes	Passive	Yes	Secondary market	Fair value
Indonesia	No	Monetary policy; Market functioning	Yes	Passive	Yes	Primary and secondary market	Face value
Korea	No	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Malaysia	No	Market functioning	No	N.A.	No	Primary and secondary market	Amortized cost
Philippines	No	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Thailand	No	Market functioning	No	N.A.	No	Secondary market	Amortized cost
China	No	Monetary policy; Market functioning	No	N.A.	No	Secondary market	Amortized cost
Serbia	No	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Hungary	Yes	Monetary policy; Market functioning	Yes	Passive	No	Primary and secondary market	Amortized cost
Croatia 3/	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Poland	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Romania	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Iceland	No	Monetary policy; Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Costa Rica	No	Market functioning	Yes	Active and passive	No	Secondary market	Amortized cost
Guatemala	No	Market functioning	Yes	Active and passive	No	Secondary market	Amortized cost
Sri Lanka	No	Monetary policy; Market functioning	Yes	Passive	No	Primary and secondary market	Face value
Jamaica	No	Market functioning	Yes	Passive	No	Secondary market	Amortized cost
Egypt	No	Market functioning	No	N.A.	No	Secondary market	Fair value
Argentina	No	Monetary financing; Market functioning	Yes	Passive	No	Primary and secondary market	Amortized cost
Ukraine	No	Monetary financing	Yes	Passive	No	Primary and secondary market	Amortized cost

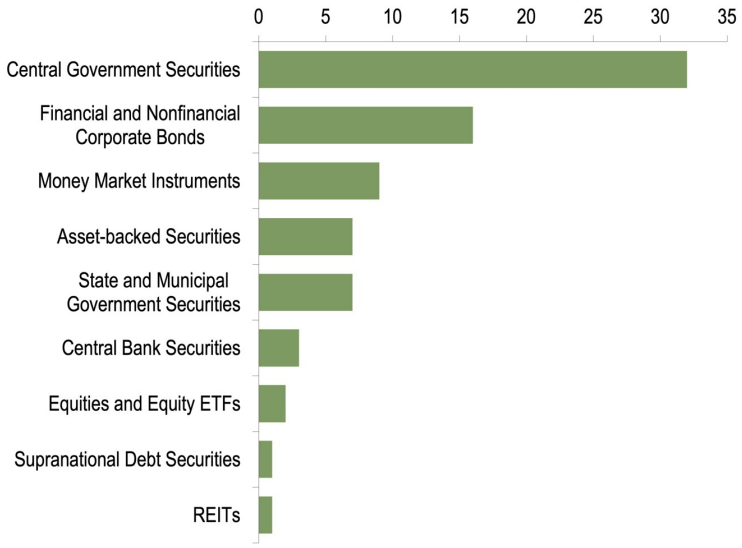
Note: This table summarizes key characteristics of central bank asset purchase programs across economies regarding data disclosure, stated objectives, operational features, market of purchase, and accounting standard. 1/ Switzerland's central bank uses issue price plus cumulative discount accretion as its accounting standard. 2/ Japan's central bank uses amortized costs for fixed income securities and moving-averages costs re-evaluated at the end of March and September for ETFs and REITs as its accounting standard. 3/ Croatia is treated as a separate economy before its euro adoption in 2023.

Table 2: APP Asset Classes by Country

Country	Central government securities	State and municipal government securities	Financial and nonfinancial corporate bonds	Asset-backed securities	Money market instruments	Equities and equity ETFs	REITs	Supranational debt securities	Central bank securities
United States	Y	Y	Y	Y	Y				
Euro Area	Y	Y	Y	Y	Y			Y	
Japan	Y		Y	Y	Y	Y	Y		
United Kingdom	Y		Y		Y				
Sweden	Y	Y	Y	Y	Y				
Canada	Y	Y	Y	Y	Y				
Switzerland			Y	Y					Y
Türkiye	Y								
Australia	Y	Y							
New Zealand	Y	Y							
South Africa	Y								
Chile			Y		Y				Y
Colombia	Y		Y						
Mexico	Y								
Israel	Y		Y						
India	Y	Y							
Indonesia	Y								
Korea	Y		Y		Y				
Malaysia	Y								
Philippines	Y								
Thailand	Y								Y
China	Y								
Serbia	Y		Y						
Hungary	Y		Y	Y					
Croatia	Y		Y						
Poland	Y		Y						
Romania	Y								
Iceland	Y		Y		Y				
Costa Rica	Y								
Guatemala	Y								
Sri Lanka	Y								
Jamaica	Y								
Egypt						Y			
Argentina	Y								
Ukraine	Y								

Note: This table summarizes different types of asset classes bought under each central bank's asset purchase programs.

Figure 2: APP Asset Class Count



Note: This figure reports the number of central banks that used each APP instrument category at least once during the sample period.

banks have ventured into major publicly traded asset classes (Table 2). Among 35 central banks that have conducted APPs in the sample from 2008 to 2024, all but three have purchased central government debt securities (Figure 2). Furthermore, 20 central banks have acquired other domestic marketable debt securities. For instance, 7 central banks have bought state and municipal government debt securities. Financial and nonfinancial corporate bonds such as agency bonds and bank bonds are the second most frequently bought assets – around half of the central banks have acquired them. Money market instruments (commercial papers, bankers' acceptances, certificates of deposit, etc.) and asset-backed securities (asset-backed bonds, mortgage-backed securities, covered bonds, etc.) are two other types of asset classes sometimes targeted by central banks. In rare instances, equities and equity ETFs and Real Estate Investment Trusts (REITs) are also included in the APP portfolios. Some National Central Banks (NCBs) in the Eurosystem have purchased bonds issued by European Union supranational institutions under the Asset Purchase Programme and the Pandemic Emergency Purchase Programme. Interestingly, central banks in certain cases have even bought back their own securities.

While APPs were mainly used by a few advanced economy central banks before 2020, many central banks in emerging markets followed suit after the COVID pandemic (Figure 3). Although the Bank of Japan's APP dated back to 2001, the first wave of APPs came after

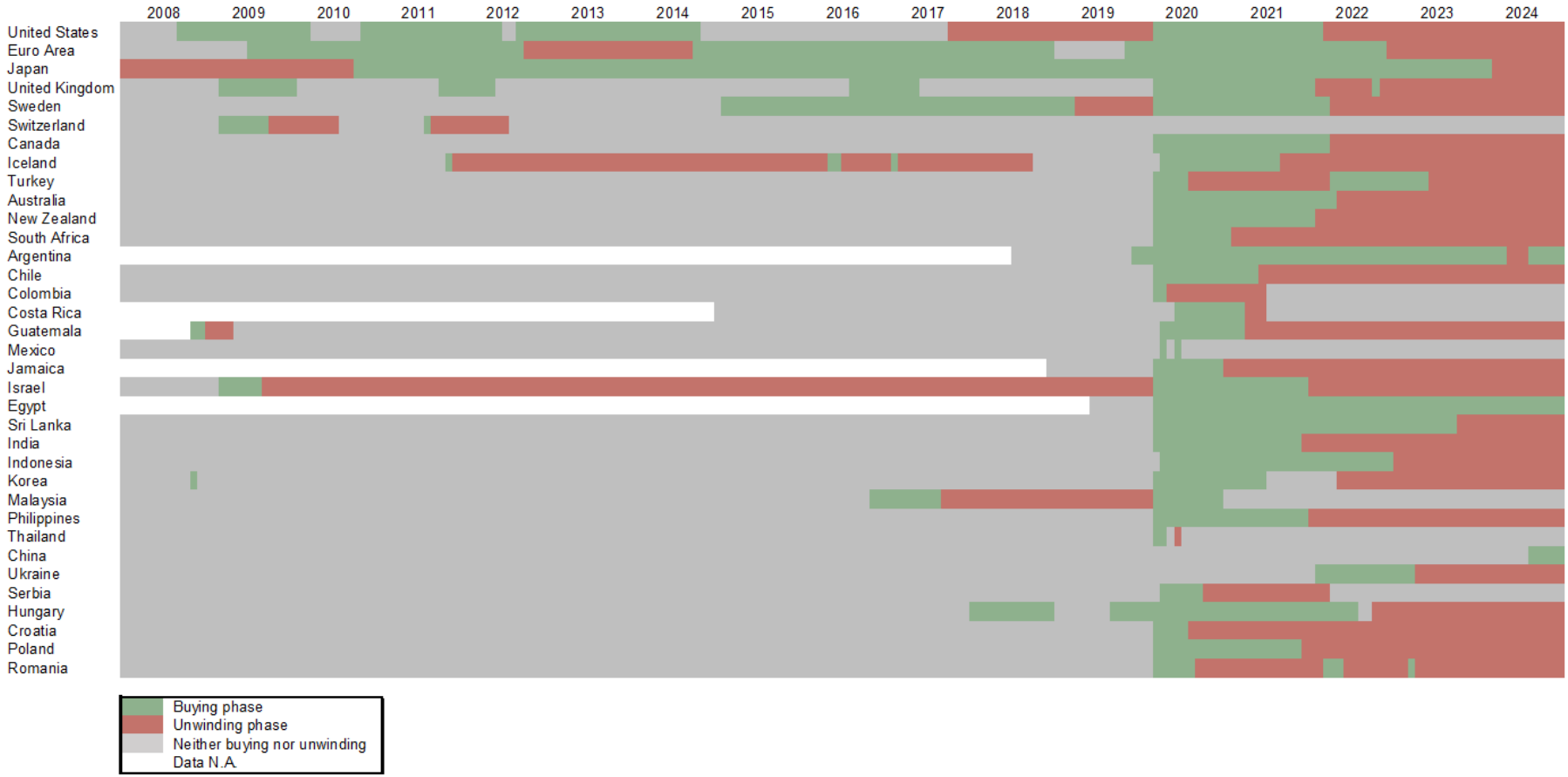
the Global Financial Crisis. The years 2008-2009 saw initiations of APPs by three systemic central banks: the Federal Reserve, the European Central Bank, and the Bank of England. Other central banks in Switzerland, Israel, South Korea, and Guatemala also embarked on asset purchase activities. Nevertheless, the total number of central banks that conducted APPs appears to be limited and they are mostly advanced economy central banks at that time. The COVID-19 pandemic was a watershed moment in the use of APPs, and the onset of the public health crisis witnessed the first truly global wave of APPs: systemic central banks reactivated their APPs, more advanced economy central banks joined forces, and many emerging market central banks followed suit. Before 2020, 8 AE central banks (60 percent of total) and 3 EM central banks (10 percent of total) had applied APPs in their monetary policy toolkit. After 2020, the total count and shares of APPs rose further in AEs and surged in EMs. During 2020-2024, 11 AE central banks (72 percent of total) and 23 EM central banks (70 percent of total) included APPs, reflecting how monetary policy practices once considered unorthodox have now become mainstream (Figure 4).

Central banks' APP amounts have varied significantly across economies, and, on average, advanced economy central banks have purchased more assets than their emerging market counterparts. For example, the peak levels of securities holdings by the Federal Reserve and the Bank of England both exceeded 30 percent of GDP. The APP portfolio held by the Bank of Japan surpassed 100 percent of Japanese GDP after 2020 (Figure 5). In contrast, several central banks, including those in Costa Rica, Malaysia, Romania, and South Africa, purchased domestic securities amounting to less than 1 percent of GDP. A direct comparison of the distribution of APP holdings across monetary authorities in AEs and EMs points to the greater propensity of AE central banks to absorb larger quantities of domestic securities from the market (Figure 6). The median values of APP holdings for AE and EM central banks were small relative to GDP before 2020. After the COVID-19 pandemic, the peak median APP holdings of AE and EM central banks rose to about 15 percent and 2 percent of GDP, respectively.

Nearly all securities acquired under APPs were purchased with the dual objectives of monetary policy and market functioning (Figure 7). By the end of 2024, purchases under programs with both monetary policy and market functioning objectives accounted for 98.6 percent of the total APP amount across all central banks. Programs with the sole objective of market functioning accounted for only 1.4 percent of the total APP amount in December 2024, up from less than 0.1 percent in December 2019.

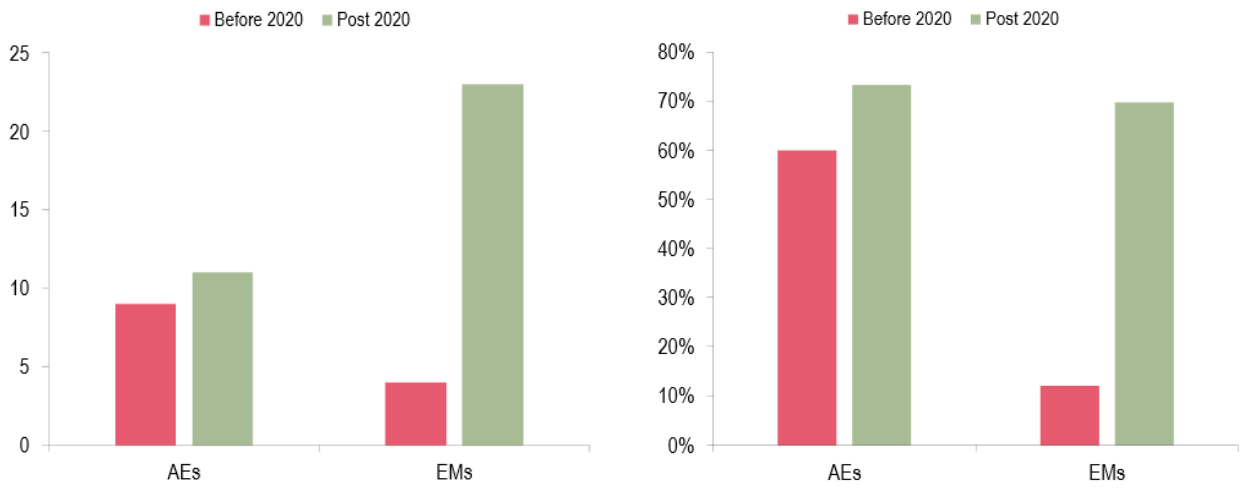
The "Big Four" central banks have dominated APPs in terms of absolute and relative sizes (Figure 8). Judging by nominal amounts, total APP amount by all central banks in the

Figure 3: APP Timeline by Country



Note: The figure plots the timeline of asset purchase program episodes across economies, distinguishing between net buying (green) and unwinding (red) phases. Central bank announcements take precedence over a mechanical cycle dating method in coding APP timelines.

Figure 4: Adoption of Asset Purchase Programs by Income Group and Time



(a) Number of Economies with APPs

(b) Share of Economies with APPs

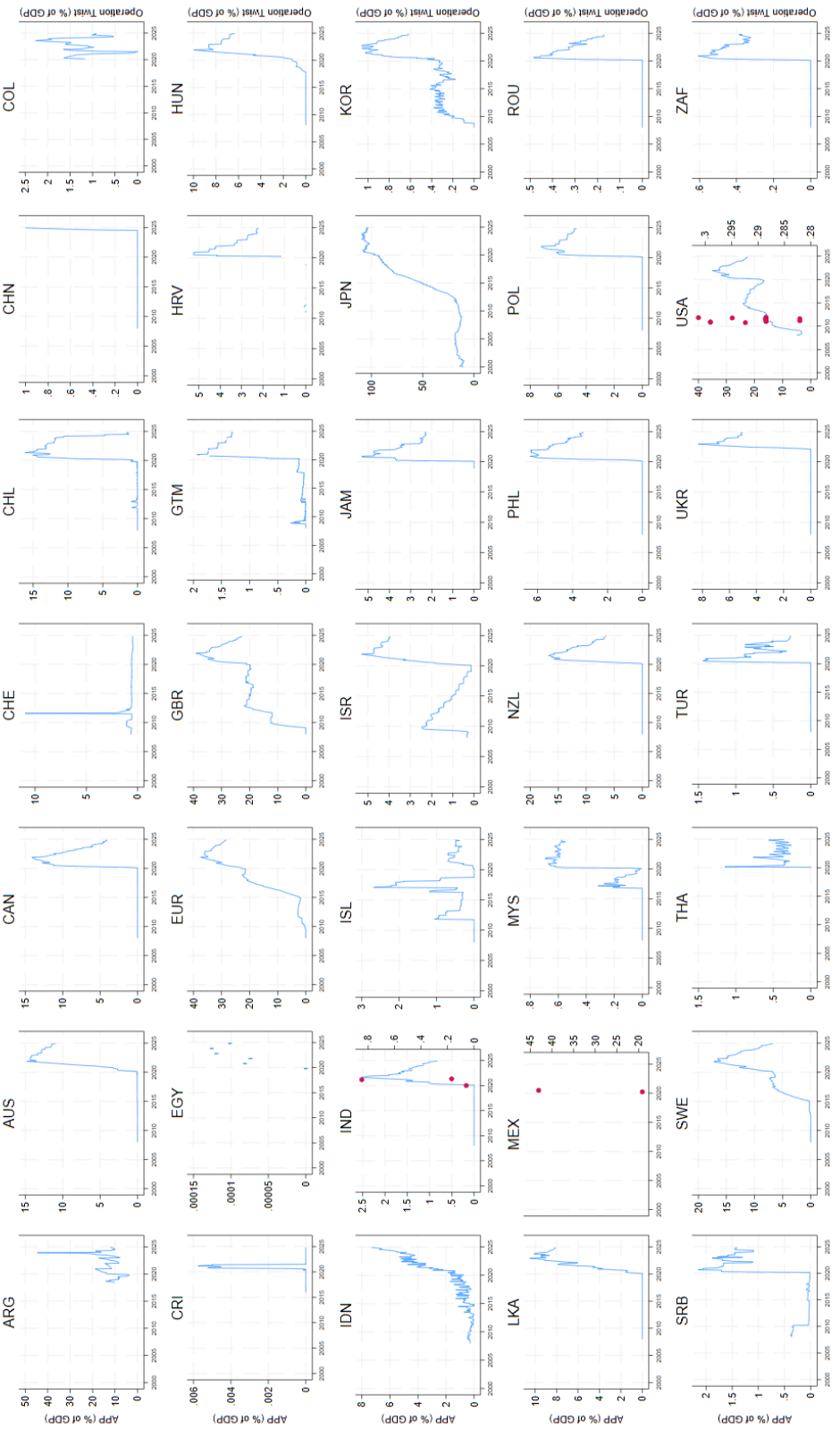
Note: Panel (a) shows the number of advanced and emerging market central banks that implemented APPs. Panel (b) reports the corresponding shares within each income group. Results are shown separately for pre-2020 and post-2020 periods.

sample reached its peak of 21.1 trillion U.S. dollars in February 2022, followed by a moderate decline to 16.6 trillion U.S. dollars in December 2024. The Federal Reserve led the APP amount, with 6.1 trillion U.S. dollars of securities holdings (35 percent of total) at end-2024. The Euro System ranked second and held 4.5 trillion U.S. dollars of securities (27 percent of total) at end-2024. The Bank of Japan stood third and accumulated 4 trillion U.S. dollars of domestic securities (24 percent of total) at end-2024. The Bank of England's APP holdings were smaller than those of its counterparts in the U.S., Eurozone, and Japan. Nevertheless, the U.K. central bank assembled 0.8 trillion U.S. dollars of domestic securities (5 percent of total) in the same period. The above "Big Four" central banks accounted for about 95 percent of total APP amounts in nominal terms at the end of 2024. The fact that these four central banks' sheer volume of APP holdings is overwhelmingly large is not surprising given their combined economic and capital market size. Other AE central banks and their EM peers piled 0.4 and 0.5 trillion U.S. dollars of domestic securities (3 percent and 2 percent of total), respectively.

General government debt securities have taken the lion's share of the APP amounts and usage count (Figure 9). General government securities, which include debt securities issued by central, state, and municipal governments, were 14.4 trillion U.S. dollars of securities (84 percent of total) at end-2024. Other domestic marketable securities took the residual share of 16 percent in December 2024.

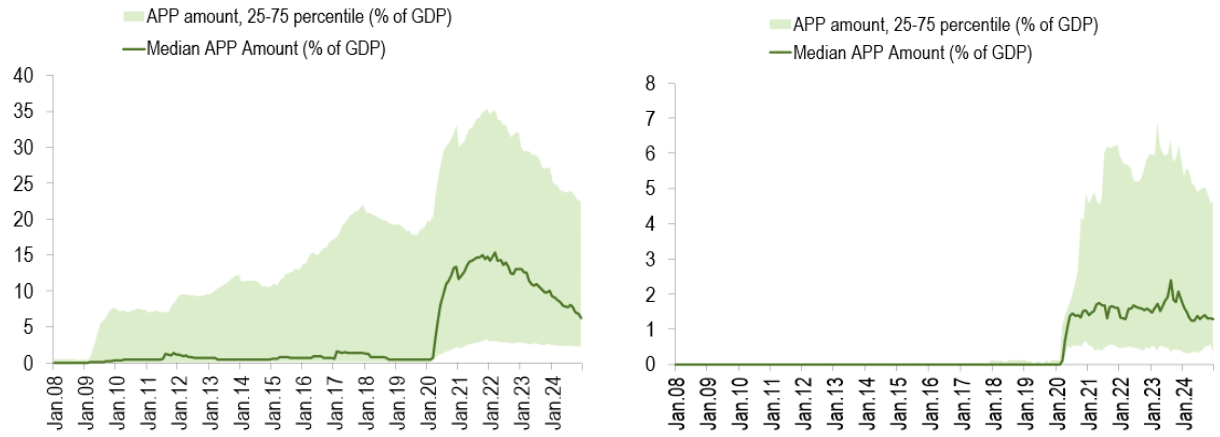
Average durations of APPs tend to last longer in advanced economies (Figure 10). Before

Figure 5: APP Amount by Country



Note: This figure plots the evolution of central bank asset purchase program (APP) amounts at the economy level, measured in percent of GDP. The blue lines show the sizes of APP holdings over time while the red dots represent the sizes of operation twists. Time series APP data are based on central banks' reported accounting standards. The vertical axes are country-specific.

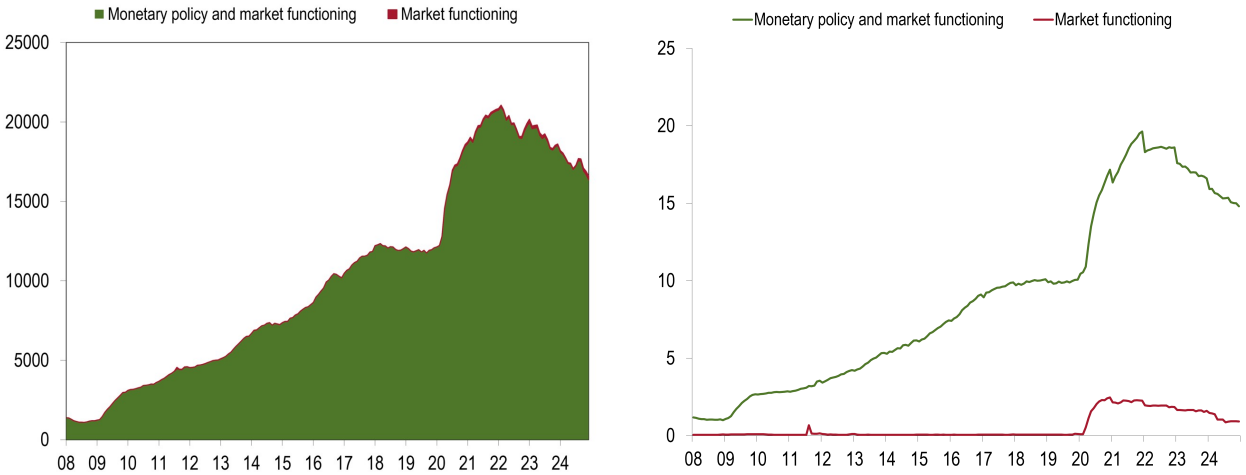
Figure 6: Distribution of Asset Purchase Program Amounts



(a) Advanced economies (b) Emerging markets

Notes: This figure shows the cross-country distribution of APP amounts over time by country income group, measured in percent of GDP. The shaded area represents the 25th–75th percentile range across countries, and the solid line represents the median. Time series APP data are based on central banks' reported accounting standards.

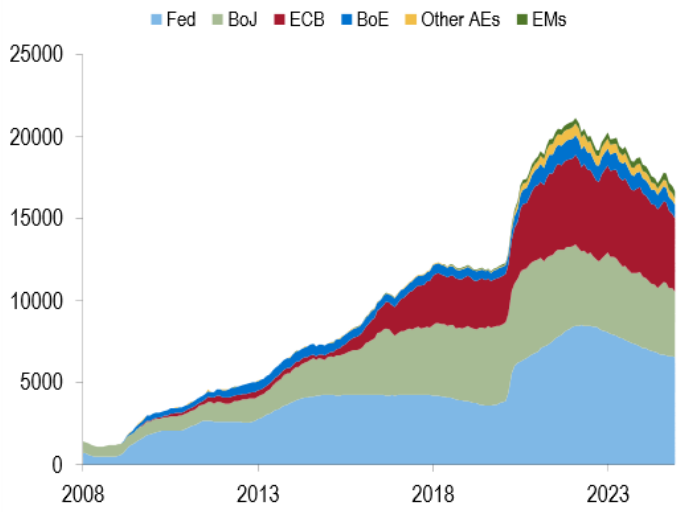
Figure 7: Asset Purchase Amounts by Program Objectives



(a) Total APP amount (b) Average APP amount as GDP percentage

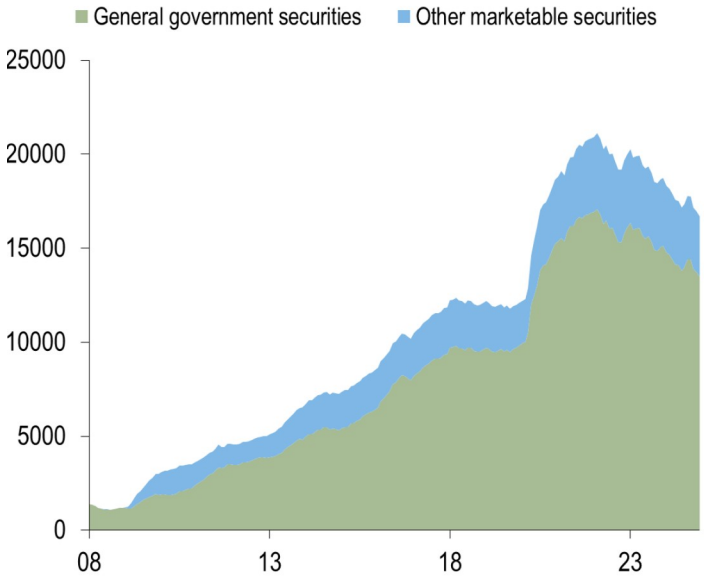
Note: The left panel shows total APP amounts by objective, while the right panel shows average APP amounts as a percentage of GDP. Programs are divided into those with a solely market functioning objective and those with both monetary policy and market functioning objectives. Time series APP data are based on central banks' reported accounting standards.

Figure 8: APP Amount by Central Bank



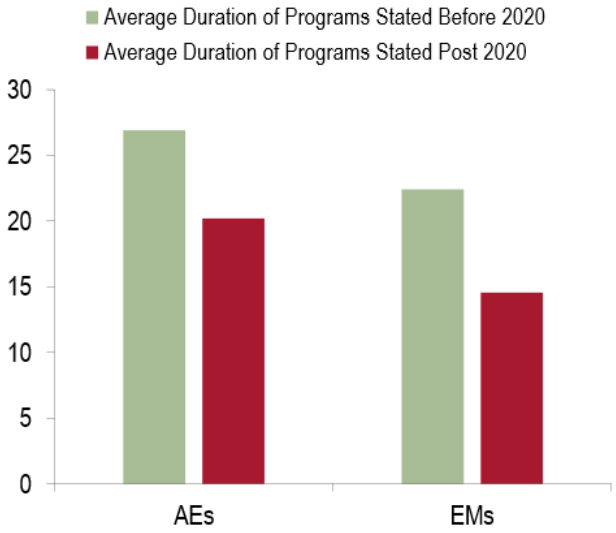
Note: This figure shows total nominal amounts of asset purchase programs (APP) by all advanced economy and emerging market central banks, measured in billions of U.S. dollars. APP amounts are constructed from program-level data by the authors based on central banks' reported accounting standards.

Figure 9: APP Amount by Asset Class



Note: This figure shows total nominal amounts of asset purchase programs (APP) by asset class across all advanced economy and emerging market central banks, measured in billions of U.S. dollars. APP amounts are constructed from program-level data by the authors based on central banks' reported accounting standards.

Figure 10: Average APP Duration



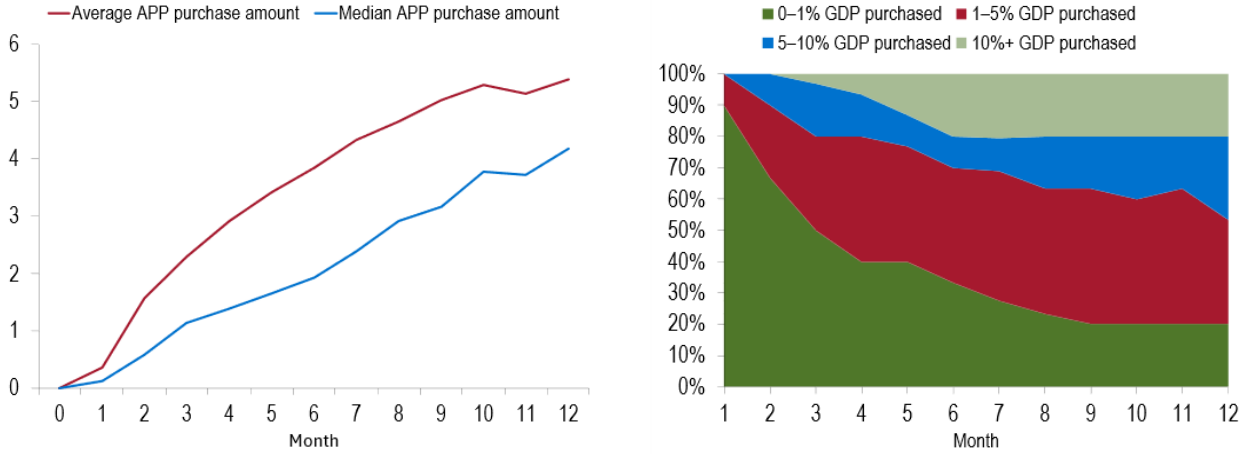
Note: This figure reports the average duration of asset purchase programs (APP) episodes in number of months by country income group across time. APP episode durations are constructed from program-level data by the authors.

2020, the average duration of the buying phase of APPs by AE central banks was 27 months, compared to 22 months by EM central banks. After 2020, more monetary authorities from smaller economies brought in APPs. The average buying phase of APPs in AEs continued for about 20 months, while EM central banks spent slightly more than one year (about 15 months) on net asset purchases.

APPs typically expand rapidly during buying phases (Figure 11). Twelve months after the beginning of an APP, average and median amounts of central bank asset purchases are equal 5.4 percent and 4.2 percent of GDP, respectively. The slope of average APPs is concave, suggesting larger buying volumes initially and reduced volumes in later stages. About 33 percent of central banks have accumulated assets worth 1–5 percent of GDP in 12 months, and another 27 percent of central banks have amassed 5-10 percent of GDP asset holdings. 20 percent of central banks have even bought assets worth more than 10 percent of GDP, while 20 percent of their peers have taken a restrained approach to limit asset purchases under 1 percent of GDP.

The unwinding of APPs is usually gradual (Figure 12). Twelve months into the unwinding phase, average and median amounts of APPs have declined by about 25 percent and 17 percent from their peak holdings at the start of portfolio reductions. About 55 percent of central banks end up unwinding 10-25 percent of their stock of APP holdings after 12 months,

Figure 11: Purchase Dynamics of Asset Purchase Programs



(a) Average and Median Purchase Pace

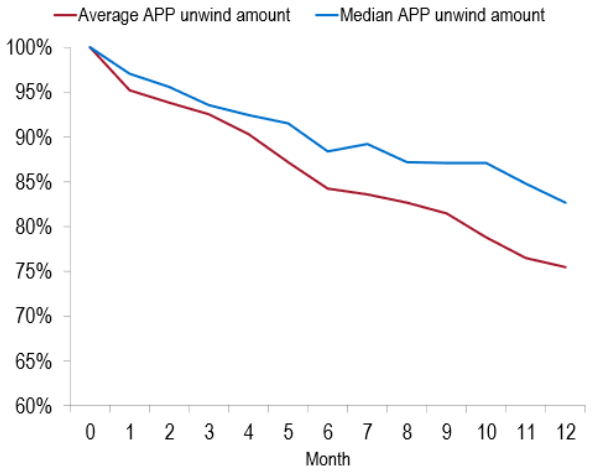
(b) Distribution of Cumulative Purchases Over Time

Note: Panel (a) summarizes the average and median pace at which central banks accumulate asset holdings during APP purchase phases, measured as cumulative purchase amounts in percent of GDP. Panel (b) shows the distribution of APP program purchase paces, expressed as a percentage of GDP, over months since the start of purchase programs.

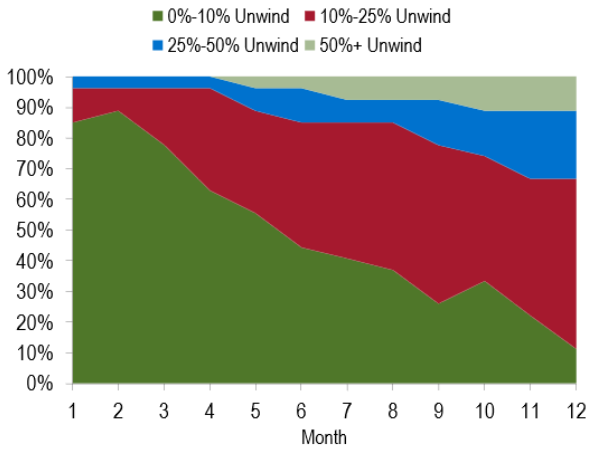
followed by 22 percent of central banks cutting their APP portfolio by 25-50 percent. Another 11 percent of central banks manage to offload more than half of the APP holdings, while the rest of the central banks cautiously lower their APP portfolio by less than 10 percent.

Central banks' policy responses around the COVID-19 pandemic have drawn interesting comparisons between policy rates and APPs (Figure 13). At the onset of the public health crisis, central banks aggressively cut policy rates. There was little policy space remaining to further lower interest rates as they came close to zero with some economies already constrained by the Zero Lower Bound. APPs, seemingly unconstrained, became an alternative policy tool to add monetary stimulus and were scaled up in 2020. Confronted with rising inflationary pressures in 2022, central banks resorted to policy rates as the main tool to tackle the problem and hiked rates well above their pre-pandemic levels. APP holdings, on the other hand, were treated more conservatively as a tool to combat inflation. Whether in AEs where central banks have accumulated much higher securities holdings or in EMs where such holdings were relatively small to economic output, the stock of APP securities holdings has declined from their post-pandemic peaks but nevertheless remained elevated above their end-2019 levels. These facts suggest that while it is easy for central banks to deploy APPs during crises, returning balance sheets to their pre-crisis sizes has proved difficult.

Figure 12: Unwinding Dynamics of Asset Purchase Programs



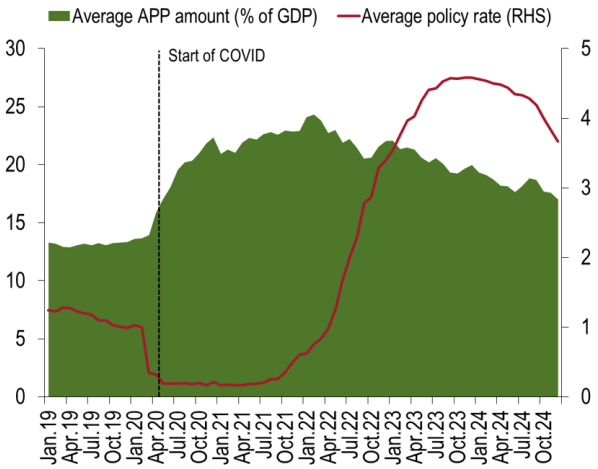
(a) Average and Median Unwind Pace



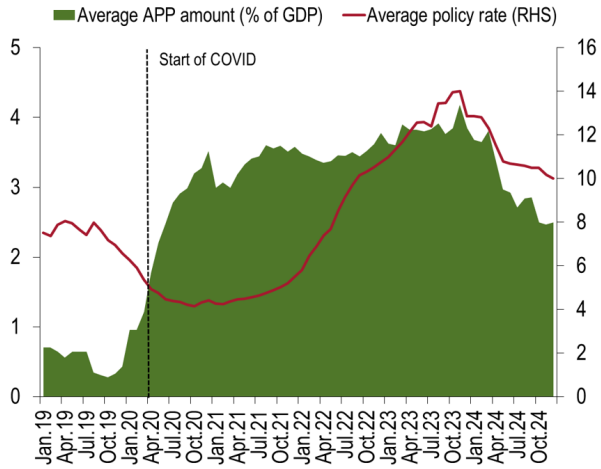
(b) Distribution of Cumulative Unwind Over Time

Note: Panel (a) summarizes the average and median pace at which central banks reduce APP holdings during unwind phases, measured as cumulative amounts in percent of peak APP holdings before the unwind. Panel (b) shows the distribution of cumulative APP portfolio reductions over months since the start of unwinding.

Figure 13: Policy Rates and APPs Before and After COVID



(a) Advanced Economies



(b) Emerging Market Economies

Note: The figure plots the evolution of policy rates and APP portfolios around the COVID period. Panel (a) shows advanced economy central banks, and Panel (b) shows emerging market central banks.

4 Empirical Analysis of Central Bank Asset Purchase Programs

In this section, we examine what factors prompt central banks to activate APPs, what initial conditions affect the duration of APPs' buying phases, and what variables determine the net purchase amounts of APPs. Our empirical analysis excludes 3 outlier countries (Argentina, Russia, and Ukraine) because of monetary financing and wars.

4.1 Triggers of Asset Purchase Programs

A central research question of this paper is why some central banks have conducted APPs while others have not, and what factors triggered central banks to start asset purchase programs. As illustrated in the previous section, APPs were used in isolated cases during the Global Financial Crisis but proliferated in the COVID-19 pandemic. While the officially stated objectives of APPs are often monetary policy or market functioning purposes, it is meaningful to unpack central banks' considerations into concrete variables, the specific channels through which they shape decisions, and shifts toward mainstream use over time.

$$\Pr(APP_i = 1) = F(X_i'\beta + \alpha_i)\,, \tag{1}$$

We run Probit regressions to estimate the probability of activating APPs by central banks. Our sample includes 45 advanced economies and emerging markets that have monetary autonomy. Given that most APPs started in the Global Financial Crisis and the COVID-19 pandemic, we focus on these two crisis periods and split the analysis accordingly. As shown in Equation (1), the dependent variable is a dummy variable that takes the value of unity if a central bank conducts APPs or zero if it does not initiate APPs. The explanatory variables include changes in real GDP growth forecast, changes in CPI inflation forecast, changes in unemployment rate forecast, sovereign bond yield volatility, a dummy variable for advanced economies, a dummy variable for the Zero Lower Bound, public debt to GDP ratio, and a financial development index from Svirydzenka (2016)[10]. Changes in macroeconomic forecasts are obtained from the IMF's World Economic Outlook (WEO) database[11], and sovereign bond

[10]One-period lagged values of the explanatory variables are used to represent the initial conditions in the analysis, except for the macroeconomic forecast revision variables.

[11]For the analysis of APP triggers during 2008-2009, changes in macroeconomic forecasts are measured as the differences between 2008 October WEO projections and 2008 April WEO projections of specific macroeconomic variables in 2009. For the analysis of APP triggers in 2020, changes in macroeconomic forecasts are measured as the differences between 2020 April WEO projections and 2019 October WEO

yield volatility is defined as the monthly standard deviation of the daily 10-year government benchmark bond yield[12] based on daily data from the secondary market.[13]

APP activations were idiosyncratic during 2008-2009. Columns (1) and (2) of Table 3 suggest that real economic growth, inflation, and government bond market volatility are not factors that lead to the initiation of APPs given that statistical insignificance. Columns (3) and (4) indicate that being an advanced economy increases the probability of activating APPs. This is not surprising, considering that 6 AEs and only 1 EM conducted APPs during 2008-2009. Furthermore, a higher level of financial development raises the probability of starting APPs. Nevertheless, a country's financial development is highly correlated with its income level. After including the dummy for advanced economies, the financial development index is no longer statistically significant. In a nutshell, we did not find any specific common factor except for an AE dummy variable essential to explaining triggers of APPs during the Global Financial Crisis.[14]

In 2020, central banks activated APPs in response to disinflationary risks and government bond market volatilities. Columns (1) through (5) of Table 4 show that the change in the inflation forecast and sovereign bond yield volatility are significant drivers of initiating APPs. Specifically, when inflation projections are revised down and government bond market volatilities surge, the chance of activating APPs rises. According to the estimation results, a decrease in the CPI inflation forecast or a rise in sovereign bond yield volatility leads to an increase in the predicted probability of activating APPs. Interestingly, expected changes in economic growth and the unemployment rate do not contribute to the likelihood of starting APPs. Moreover, advanced economy central banks do not enjoy a higher probability of starting APPs as 11 AE and 20 EM central banks conducted APPs in 2020. Central banks that are constrained by the Zero Lower Bound also do not demonstrate a greater probability of initiating APPs. Finally, Columns (6) through (9) suggest that when deciding whether to activate APPs or not, central banks in advanced economies did not differ from their emerging market peers. None of the interaction terms between AE central bank dummy and

projections of specific macroeconomic variables in 2020.

[12]Since we aim to capture disorderly conditions in domestic markets, our default measure of 10-year government benchmark bond yield is the local currency bond yield. However, such data are not available for 7 countries (Albania, Armenia, Costa Rica, Guatemala, Paraguay, Serbia, and Uruguay). Therefore, we use external government bond yields for these countries instead. For the Euro Area, the measure is monthly standard deviation of GDP-weighted 10-year government bond daily yields of 5 Eurozone periphery countries (Greece, Italy, Ireland, Portugal, and Spain).

[13]A better measure of government bond market liquidity is the abundance or shortage of arbitrage capital, which eliminates big price deviations from fundamental values. Hu, Pan, and Wang (2013) analyze the noise in the price of U.S. Treasuries and examine its informativeness as a measure of overall market liquidity

[14]We are unable to include change in the unemployment rate forecast in this part of the analysis due to data availability.

Table 3: Triggers of Asset Purchase Programs: 2008–2009

	(1)	(2)	(3)	(4)	(5)	(6)
Change in 2009's real GDP growth forecast	-0.105	-0.586	0.195	0.124	-0.002	0.160
	(0.186)	(0.412)	(0.310)	(0.564)	(0.193)	(0.304)
Change in 2009's CPI inflation forecast	-0.276*	-0.223	-0.0597	-0.0338	-0.107	-0.0419
	(0.149)	(0.149)	(0.0945)	(0.164)	(0.0784)	(0.0911)
10-year sovereign bond yield volatility		-0.0740		-0.0558		
		(0.0559)		(0.0871)		
Advanced economy dummy			1.598**	1.356*		1.431**
			(0.634)	(0.777)		(0.701)
General government debt to GDP, lagged					-0.00442	-0.00624
					(0.00666)	(0.00663)
Financial development index, lagged					2.559**	0.729
					(1.260)	(1.611)
Constant	-0.755**	-0.389	-1.612***	-1.053	-2.139**	-1.692*
	(0.304)	(0.541)	(0.465)	(0.936)	(0.896)	(0.944)
Pseudo R^2	0.0802	0.1273	0.2741	0.2239	0.2291	0.2914
Observations	45	34	45	34	45	45

Note: The table reports probit regressions estimating the probability of initiating an asset purchase program (APP) during 2008–2009. The estimated specification is $\Pr(APP_i = 1) = F\left(X_i'\beta + \alpha_i\right)$, where APP_i is a dummy variable equal to one if country i initiates an APP in 2008-2009, where X is a vector of macroeconomic and financial variables. Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

macroeconomic and financial factors are statistically significant, pointing to no difference between the reaction functions of AE and EM central banks in commencing APPs.

4.2 Determinants of Asset Purchase Program Durations

Another key research question is how long APPs last and what initial conditions affect the length of APPs. Our focus in this study is on the durations of APP buying phases, because unwinding phases often end when new crises such as the COVID-19 pandemic emerge and most APPs activated in 2020 were still in unwinding phases and unfinished by the end of our sample period (i.e., December 2024). All but 4 buying phases of APPs in our sample are not right-censored and offer nearly perfect observations for estimating true statistical parameters in survival analysis.

Table 5 presents statistics on all APP episodes in our sample covering the period of 2008-2024. In total, there are 57 APP episodes by 12 AE and 21 EM central banks. The buying phases of APPs have a median buying duration of 1 year and a mean buying duration of 1.63 years. A few APP episodes have long purchase cycles, as suggested by the large standard deviation and the difference between the mean and the median. AE central banks not only have more frequent APP episodes, but they also tend to carry buying phases of APPs into more extended times. The median (mean) values of buying phase durations are

Table 4: Triggers of Asset Purchase Programs: 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Change in 2020's real GDP growth forecast	-0.124	-0.138	-0.152	-0.163	-0.190	-0.156	-0.149	-0.153	-0.171
	(0.110)	(0.111)	(0.117)	(0.137)	(0.122)	(0.134)	(0.136)	(0.118)	(0.134)
Change in 2020's CPI inflation forecast	-0.564***	-0.544***	-0.559***	-0.683**	-0.616**	-0.476**	-0.471**	-0.677**	-0.581**
	(0.205)	(0.217)	(0.228)	(0.275)	(0.248)	(0.231)	(0.230)	(0.305)	(0.268)
10-year sovereign bond yield volatility		0.110**	0.118**	0.143**	0.134**	0.128**	0.127**	0.162***	0.145**
		(0.0524)	(0.0527)	(0.0597)	(0.0581)	(0.0557)	(0.0558)	(0.0595)	(0.0588)
Advanced economy dummy			-0.260	-0.907	-1.083				
			(0.546)	(0.542)	(0.943)				
Zero lower bound dummy			0.246				-0.636		
			(0.857)				(0.994)		
Change in 2020's unemployment rate forecast				0.0512				0.412	
				(0.139)				(0.259)	
General government debt to GDP, lagged					0.00545				0.00430
					(0.00643)				(0.00757)
Financial development index, lagged					2.015				1.695
					(1.986)				(1.899)
Change in 2020's real GDP growth forecast × AE dummy						-0.0204	-0.0628	-0.152	-0.00210
						(0.155)	(0.174)	(0.174)	(0.678)
Change in 2020's CPI inflation forecast × AE dummy						-0.700	-1.040	-0.364	-0.0542
						(0.701)	(0.856)	(0.855)	(0.678)
10-year sovereign bond yield volatility × AE dummy						-0.109	-0.157	-0.0608	-0.122
						(0.149)	(0.151)	(0.163)	(0.163)
Change in 2020's unemployment rate forecast × AE dummy								-0.525	
								(0.348)	
Constant	-0.675	-1.579*	-1.653*	-2.260**	-3.150**	-1.780*	-1.677*	-2.908***	-2.911***
	(0.814)	(0.913)	(0.923)	(1.032)	(1.348)	(0.964)	(0.976)	(1.071)	(1.083)
Pseudo R^2	0.1414	0.1902	0.1947	0.2512	0.2502	0.2156	0.2214	0.3453	0.2473
Observations	45	43	43	40	43	43	43	40	43

Note: The table reports probit regressions estimating the probability of initiating an asset purchase program (APP) in 2020. The estimated specification is $\Pr(APP_i = 1) = F\left(X_i'\beta + \alpha_i\right)$, where APP_i is a dummy variable equal to one if country i initiates an APP in 2020, where X is a vector of macroeconomic and financial variables. Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

1.67 and 0.83 (2.05 and 1.19) years for AE and EM central banks, respectively.

Table 5: Summary Statistics of Asset Purchase Program Episodes

Sample	# Episodes	# Countries	Episodes per Country	Median Duration	Mean Duration	Std. Dev.
All	57	33	1.73	1.00	1.63	2.07
AEs	29	12	2.42	1.67	2.05	2.56
EMs	28	21	1.33	0.83	1.19	1.32

Note: This table reports summary statistics of asset purchase programs' episodes and durations. Duration is measured in years. Advanced economies (AEs) and emerging markets (EMs) are classified according to the IMF World Economic Outlook definition. Argentina and Ukraine are excluded in the analysis.

The distribution of APP buying phase durations concentrates on short elapses of time but possesses long tails (Table 6). More than half of all APPs' buying phases end within one year, but 2 episodes by the Bank of Japan last beyond 5 years. Durations of 0–1 and 1–2 years are both prevalent among AE central banks' APPs, and 10 additional episodes continue asset purchases for more than two years. Buying phases of APPs by EM central banks are predominantly short-lived, as 19 out of 28 episodes stop asset purchases in less than a year. As seen in our sample statistics, no single buying spell of APPs by EM central banks exceeds 5 years.

Table 6: Distribution of Asset Purchase Program Buying Phase Durations

Sample	0–1	1–2	2–3	3–4	4–5	5–6	>6
All	30	13	5	3	4	1	1
AEs	11	8	4	2	2	1	1
EMs	19	5	1	1	2	0	0

Note: This table reports the number of APP episodes by duration interval (in years). Duration of each episode's buying phase is based on central bank announcements when available or an identification algorithm discussed in the paper. Argentina and Ukraine are excluded in the analysis.

We conduct survival analysis[15] to examine survival rates, duration dependence, and initial factors that affect duration dependence of APPs' buying phases. The statistics in Table 7 show unconditional survival probabilities of APP buying phases are higher in AEs than in EMs, in line with the distribution of APP buying phases discussed above. Around three quarters of all APP buying spells do not last more than two years. The conditional probability that a current APP purchase cycle will continue in the subsequent periods does not reveal monotonic patterns over time. The persistence of maintaining asset purchases is fairly strong for both AE and EM central banks. For instance, 56% and 44% of AE and EM central banks that have asset purchases in the second year continue buying spells in the third year.

Table 7: Survival Probabilities of Asset Purchase Program Buying Phases

Panel A. Unconditional Probabilities					
Sample	>1 year	>2 years	>3 years	>4 years	>5 years
All	48%	25%	16%	11%	4%
AEs	62%	34%	21%	14%	7%
EMs	32%	14%	11%	7%	N/A
Panel B. Conditional Probabilities					
Sample	>1 year	>2 years	>3 years	>4 years	>5 years
All	48%	52%	64%	67%	33%
AEs	62%	56%	60%	67%	50%
EMs	32%	44%	75%	67%	N/A

Note: Panel A reports unconditional survival probabilities that an asset purchase program's buying phase lasts longer than a given year. Panel B reports conditional survival probabilities that an asset purchase program's buying phase lasts beyond year t, conditional on its survival probability through year $t-1$.

We apply the Cox proportional hazards model for survival analysis.[16] The Cox proportional

[15]Survival analysis is a standard method in medical research and has many applications such as evaluation of how long a patient will survive after a medication or a treatment. In macroeconomics, researchers have utilized it to study how long a regime will last, when a sovereign will default, or when a country will regain international market access. For example, Klein and Shambaugh (2008) rely on survival analysis to investigate durations of fixed and floating exchange rate regimes.

[16]We code our survival dataset using a single observation per APP episode and record the beginning and

hazards regression model (Cox 1972) is a popular semiparametric model for the analysis of survival data. It is often used to assess predictors of the time to a particular event with the following hazard function:

$$h_i(t \mid X_i) = h_0(t) \exp(X\beta), \tag{2}$$

where $h_i(t \mid X_i)$ is the hazard rate at time t under a set of covariates X including a dummy variable for the Zero Lower Bound, a dummy variable for the APP objective[17], the buying duration of the last APP, public debt to GDP ratio, a financial development index, and a dummy variable for advanced economies. The advantage of the Cox model is that while the regression parameters β are estimated parametrically, no specification is imposed on the baseline hazard function $h_0(t)$. Our choice of the Cox model over other common parametric models such as the Weibull hazard function also reflects the fact that conditional survival probabilities of APP buying phases are not strictly monotonic in terms of duration dependence.

Durations of APP buying phases depend on whether the central bank is constrained by the Zero Lower Bound and the country's public debt to GDP ratio at first. Table 8 reports the estimated exponentiated hazard ratios[18] of ending asset purchases for a set of initial conditions at the beginning of APPs. Our estimates in Column (7) show that the probability that a central bank constrained by the ZLB at APP onset stops asset purchases in a given period is only 35% of the probability of a central bank not constrained by the ZLB, ceteris paribus. In addition, a one percentage point increase in the public debt to GDP ratio at the time of APP activation leads to a 1.2 percent lower likelihood (i.e., 1-0.988=1.2%) for a central bank to end asset purchases, all else equal. On the other hand, APP objective, the duration of the last APP's buying phase[19], financial development level, and an advanced economy status do not explain durations of buying cycles in APPs. To visualize the findings, Figure 14 plots survival functions for APP buying phases by our measures of monetary policy rate space and fiscal space. It is evident that central banks under the ZLB with limited policy rate space resort to APPs to put downward pressure on long-term interest rates by extending the durations of asset purchases. We also sort our sample based on the distribution of public debt ratios and pick the 25th, median, and the 75th percentiles for illustrative purposes. For

ending times of each episode. The initiation of an APP is when an episode begins and the episode ends in either a failure (i.e., conclusion of the buying phase) or a nonfailure (i.e., right censoring, as some APPs' buying phases have lasted beyond our sample period.).

[17]The dummy variable is equal to one if the stated program objective is purely for market functioning and zero otherwise

[18]A hazard ratio above one indicates an increased risk of ending an APP's buying phase, while a hazard ratio below one suggests a higher chance of continuing an APP's buying phase.

[19]We set duration of last APP's buying phase to zero if a central bank has never done an APP before.

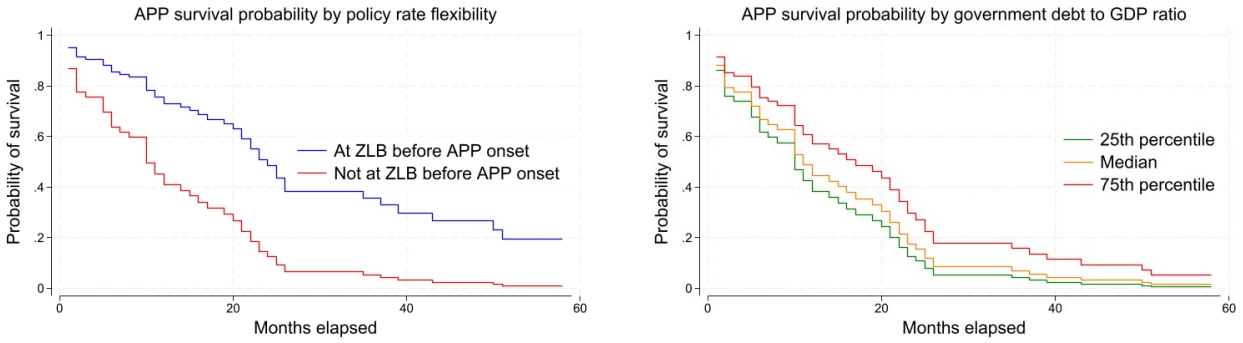
a central bank situated in a high debt country, its tendency to continue asset purchases under an APP is much greater compared with a counterpart situated in a low debt country.

Table 8: Determinants of Asset Purchase Program Buying Phase Duration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Zero Lower Bound dummy	0.337***	0.422**	0.471**	0.456**	0.366***	0.380***	0.350***
	(0.320)	(0.359)	(0.316)	(0.329)	(0.295)	(0.328)	(0.335)
APP market functioning objective dummy		1.768					
		(0.439)					
General government debt to GDP			0.990*	0.989*	0.988**	0.988**	0.988**
			(0.0061)	(0.0062)	(0.0060)	(0.0055)	(0.0058)
General government debt to GDP × market functioning dummy			1.004	1.004	1.006	1.007	1.006
			(0.0076)	(0.0075)	(0.0076)	(0.0079)	(0.0079)
Duration of last APP buying phase				1.023			1.025
				(0.0637)			(0.0686)
Financial development index					3.312		2.835
					(0.783)		(1.524)
Advanced economy dummy						1.630	1.100
						(0.349)	(0.687)
Observations	57	57	57	57	57	57	57

Note: The table reports regression results on initial conditions that determine asset purchase programs' buying phase durations. The estimates are based on a Cox proportional hazard model specified as $h_i(t \mid X_i) = h_0(t) \exp(X\beta)$, where $h_i(t)$ denotes the hazard rate of ending an APP's buying phase at duration t, $h_0(t)$ is the baseline hazard, and X includes variables at the time of APP initiation. Coefficients are reported as exponentiated hazard ratios, with robust standard errors in parentheses. A hazard ratio below one implies a longer buying phase duration for an APP. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 14: Survival Functions of Asset Purchase Program Buying Phases



(a) Zero Lower Bound Constraint

(b) Public Debt-to-GDP

Note: The figure shows estimated survival functions for APP buying phases. Panel (a) compares central banks constrained by the zero lower bound with those that are not constrained. Panel (b) compares central banks in economies with various percentiles of public debt-to-GDP ratios.

4.3 Determinants of Asset Purchase Program Paces

Our final research question is to investigate determinants of APP purchase amounts. As noted in the stylized fact section, APP amounts have varied substantially: AE central banks have accumulated more securities holdings than their EM peers, but there is considerable heterogeneity within country income groups. APPs were upsized significantly during the

COVID-19 pandemic than in prior programs, and overall purchase paces are rapid while unwinding paces are gradual. Central banks often set predetermined amounts of purchases or unwinding at the times of program initiation or sometime before actual unwinding actions take place. However, they retain sufficient discretion in program implementation and adjust the paces of APPs flexibly according to economic and financial developments. These facts motivate us to assess whether common macroeconomic and financial factors as well as past paces determine the amount of securities acquired under APPs.

We deploy dynamic panel data models to explore determinants of paces of APPs. We estimate the following dynamic panel equation:

$$y_{it} = \eta y_{i,t-1} + \boldsymbol{\beta X} + \alpha_i + \theta_t + \varepsilon_{it}, \tag{3}$$

where the dependent variable y_{it} is the net securities purchase amount (in percent of GDP) of country i in month t. η is the estimated coefficient on the lagged dependent variable, which allows for some persistence in APP paces along a predetermined path set by the central bank. X is a vector of explanatory variables to model a central bank's data-dependent behavior. It includes real GDP growth, CPI inflation, unemployment rate, volatility of the 10-year sovereign bond yield, the lagged public debt to GDP ratio, a financial development index, a dummy variable for advanced economies, and a dummy variable for the Zero Lower Bound. Country and time fixed effects are also included. We split our sample into buying and unwinding phases, as evidence presented in the earlier part of this paper suggests that patterns of central bank balance sheet expansions differ from those of balance sheet normalization. To address estimation bias in dynamic panel data regressions, the Generalized Method of Moments (GMM) is applied. The lagged dependent variable is the GMM-style instrument and time dummy variables are used as IV-style instruments.

Results suggest that the APP pace during buying phases is determined by its pace in the last period and real economic growth (Table 9). The estimated coefficient of the lagged dependent variable is highly significant and around 0.3, indicating that about 30 percent of the net purchase amount in the last period remains in the net purchase amount of an APP in the current period besides other contributing factors. The statistical significance of the lagged dependent variable is not surprising as central banks usually follow a partially mechanical approach in setting the paces of APPs. Furthermore, the estimated coefficient on real GDP growth is statistically significant and approximately -0.02. This implies that economic fundamentals matter in the calibration of APP paces, and a one percentage point decrease in real GDP growth leads to an increase in monthly asset purchases equivalent to

0.02 percent of GDP. Other control variables, however, do not offer explanatory power. The fact that sovereign bond yield volatility does not influence the paces of APPs is probably because disorderly market conditions are usually temporary and central banks have calmed down markets promptly when crises arrive.

Table 9: GMM Estimation of Paces of APPs during Buying Phases

	(1)	(2)	(3)	(4)	(5)	(6)
Net purchase amount, lagged	0.310***	0.321***	0.312***	0.291***	0.292**	0.305**
	(0.114)	(0.106)	(0.109)	(0.105)	(0.113)	(0.135)
Real GDP growth	-0.0179***	-0.0181**	-0.0183***	-0.0139	-0.0182***	-0.0175*
	(0.00674)	(0.00732)	(0.00666)	(0.0172)	(0.00649)	(0.00950)
CPI inflation	-0.0127	-0.00420	-0.0132	-0.00370	-0.0138	-0.0124
	(0.00793)	(0.00709)	(0.00860)	(0.00577)	(0.00849)	(0.00887)
Advanced economy dummy	0.111	0.138	0.0728	0.274	0.209	0.119
	(0.226)	(0.160)	(0.195)	(0.172)	(0.391)	(0.249)
10-year sovereign bond yield monthly volatility		-0.182				
		(0.368)				
General government debt to GDP ratio, lagged			0.000336			
			(0.00134)			
Unemployment rate				0.0176		
				(0.0368)		
Financial development index					-0.110	
					(0.708)	
Zero lower bound dummy						0.0150
						(0.260)
Constant	0.300***	0.283***	0.292**	0.0757	0.332	0.285
	(0.0986)	(0.0736)	(0.126)	(0.270)	(0.315)	(0.239)
Observations	987	987	987	961	987	987
Number of countries	30	30	30	29	30	30
Number of instruments	19	19	19	19	19	19
Arellano-Bond AR(1) test, p value	0.109	0.100	0.105	0.127	0.122	0.117
Arellano-Bond AR(2) test, p value	0.322	0.309	0.323	0.308	0.316	0.323
Hansen identifying restrictions test, p value	0.669	0.752	0.599	0.895	0.675	0.621

Notes: This table reports System GMM regression estimates for APP amounts during buying phases. The estimated model is $y_{it} = \eta y_{i,t-1} + \boldsymbol{\beta X} + \alpha_i + \theta_t + \varepsilon_{it}$. The dependent variable is APP net purchase amount in percent of GDP for central bank i in month t. Country and time fixed effects are included. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

During unwinding phases, the APP pace is determined by its own pace in the previous period and inflation (Table 10). Once again, the lagged dependent variable is statistically significant, which reflects that central banks typically unwind their asset holdings in a gradual and predetermined manner to avoid policy uncertainty and market panics. The estimated coefficient of the lagged dependent variable indicates a stable pace in asset holding reductions. In addition, economic fundamentals also matter during unwinding phases. The estimated coefficient on CPI inflation is statistically significant and around -0.01, meaning that a one percentage point increase in CPI inflation causes a decrease (increase) in monthly net asset purchases (reductions) equivalent to 0.01 percent of GDP. The other control variables do not affect the paces of APPs. Central banks' asymmetric reactions to economic growth

and inflation during buying and unwinding phases are interesting, and reveal that balance sheet tightening is not simply balance sheet easing in reverse in terms of asset purchase reaction functions. These findings predominantly reflect the experience around the COVID-19 pandemic as most APP episodes occurred after December 2019: central banks intensified asset purchases to bolster sluggish economic growth when COVID-19 hit the world economy in 2020 and were forced to accelerate balance sheet reductions when inflation spiked in 2022.

Table 10: GMM Estimation of Paces of APPs during Unwinding Phases

	(1)	(2)	(3)	(4)
Net purchase amount, lagged	-0.195**	-0.192**	-0.210**	-0.199**
	(0.0934)	(0.0899)	(0.0821)	(0.0854)
Real GDP growth	0.00376	0.00375	0.00183	0.00371
	(0.00530)	(0.00610)	(0.00214)	(0.00603)
CPI inflation	-0.00998*	-0.00896*	-0.00956**	-0.0108*
	(0.00521)	(0.00490)	(0.00392)	(0.00563)
Advanced economy dummy	-0.0838	-0.0808*	-0.0223	-0.0750
	(0.0639)	(0.0439)	(0.0896)	(0.0769)
General government debt to GDP ratio, lagged		0.000318		
		(0.000436)		
Unemployment rate			0.00489	
			(0.00851)	
Financial development index				-0.0515
				(0.292)
Constant	0.0188	-0.0101	-0.0341	0.0491
	(0.0370)	(0.0436)	(0.0962)	(0.125)
Observations	1,195	1,195	1,162	1,195
Number of countries	29	29	28	29
Number of instruments	19	19	19	19
Arellano-Bond AR(1) test, p value	0.144	0.132	0.149	0.137
Arellano-Bond AR(2) test, p value	0.131	0.132	0.109	0.122
Hansen identifying restrictions test, p value	0.686	0.645	0.906	0.646

Notes: This table reports System GMM regression estimates for APP amounts during unwinding phases. The estimated model is $y_{it} = \eta y_{i,t-1} + \boldsymbol{\beta X} + \alpha_i + \theta_t + \varepsilon_{it}$. The dependent variable is APP net purchase amount in percent of GDP for central bank i in month t. Country and time fixed effects are included. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Discussion and Conclusion

This paper makes a twofold contribution to the economics literature. It presents a comprehensive database of central bank APPs across the entire universe of AEs and EMs that have monetary autonomy. Our evidence shows that most central banks have conducted APPs,

which have expanded dramatically since the COVID-19 pandemic, exhibited heterogeneous characteristics, and entered into every major asset class. Furthermore, our empirical findings suggest that the triggers, durations, and paces of APPs can be explained by conventional macroeconomic and financial variables.

Several key differences between AE and EM central bank asset purchase programs are noteworthy. Before 2020, APPs were implemented primarily by AE central banks, but many EM central banks adopted them following the COVID-19 pandemic. Although all AE central banks purchased publicly traded securities in secondary markets, some EM central banks also made purchases in primary markets because of limited secondary market depth and liquidity. Compared with their EM counterparts, AE central banks purchased larger amounts of domestic securities, conducted purchases more frequently, and maintained their buying phases for longer durations. Moreover, at the onset of the pandemic, conventional monetary policy space in AEs had largely been exhausted, with policy rates at the Zero Lower Bound, whereas EM central banks did not reduce their policy rates to zero.

The slow unwinding of APP portfolios is thought-provoking. The last two decades of experiences of central bank asset purchase programs indicate that they are easy to expand but hard to fully unwind, especially in a short period of time. During our sample period from 2008 to 2024, no central bank fully unwound its APP securities holdings once these holdings exceeded 5 percent of GDP. Recent experiences also draw a sharp contrast between policy rates and APPs as monetary policy tools: Policy rates can be adjusted upward or downward quickly but are constrained by the effective lower bound. On the other hand, the upper limit on the size of APP portfolios is seemingly high, but APP portfolios are difficult to unwind. Possible reasons for slow unwinding paces, in addition to economic growth and inflation considerations, include fiscal dominance and addiction to monetary financing, worries about a resurgence in bond market volatilities, fears of a negative impact on bank reserve balances and financial system liquidity because of central bank asset sales, and concerns about realizing losses on APP securities holdings and adverse consequences for central bank capital positions.

Our database compilation process clearly indicates substantial room to improve central bank transparency and communications. The quality and quantity of historical information varied across central banks, and we did our best to locate all possible official sources on each and every APP. Few central banks have fully disclosed their APP data, even at the aggregate level. Publications of APP holdings at the security level are rare. In addition, public communications about APPs are often limited and insufficient, especially in late accumulation stages and unwinding periods of APPs. While many central banks have made loud and clear press releases upon APP activations, details are lacking in subsequent buying

periods and unwinding periods.

There are numerous use cases of our APP database for academic researchers and policymakers. For instance, the database can be used to construct asset purchase program shocks or surprises based on central bank announcements and actual amounts of asset purchases. The database comes in handy for analyzing effects of APPs beyond high-frequency event studies, especially for investigating the impact of emerging market central banks' own APPs. For theoretical monetary economists, our database and findings are timely guideposts for modeling central bank asset purchase reaction functions. A remarkable share of the 21.1 trillion U.S. dollars securities holdings is government debt, much of which was bought by central banks when interest rates were low. The database makes it feasible to examine the financial impact of APPs on central banks' capital positions and quantify the valuation gains or losses due to APP securities holdings under mark-to-market accounting rules. Lastly, our work provides a more nuanced analysis of the sovereign debt investor base by distinguishing central banks' securities holdings between APPs and conventional monetary policy operations.[20]

There are several possible extensions of our study. First, this analysis can be extended to test whether central banks demonstrate distinct behaviors when purchasing government debt securities versus other asset classes. Second, one can further examine central bank asset purchases in relation to monetary policy frameworks, output gaps, and deviations of inflation from central bank targets. Do inflation targeting central banks conduct APPs differently compared with central banks with other nominal anchors? In inflation targeting countries, how do central banks take into account actual inflation with respect to their inflation targets, target bands, and track records?[21] Third, it would be useful to explore how APPs interact with other central bank intervention measures such as liquidity and funding facilities to support the economy and preserve financial stability. We leave these questions for future research.

[20]Historically, Arslanalp and Tsuda (2012 and 2014) and Fang, Hardy, and Lewis (2025) study sovereign debt holdings by investor composition. However, prior research work does not explore central bank holdings of sovereign debt based on different motives. This paper is a step forward in understanding central banks as an increasingly important type of sovereign debt holders and their investment behaviors under various causes.

[21]For example, Zhang (2025a) offers a comprehensive assessment of 41 inflation targeting countries from 1990 to 2024 regarding their inflation targets, bands, and track records. Zhang (2025b) documents a new link between inflation targeting, markets and economic sectors and shows that inflation targeting track records have lasting impacts on financial markets. Zhang and Wang (2022) compare inflation targeting with other monetary policy frameworks and investigate the effects of inflation targeting on macroeconomic performance.

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Appendix

Table 11: Sample Country List

Countries that have conducted APPs during 2008-2024: Advanced Economies: Australia, Canada, Euro area, Iceland, Israel, Japan, Korea, New Zealand, Sweden, Switzerland, United Kingdom, United States. Emerging Markets: Argentina, Chile, China, Colombia, Costa Rica, Croatia, Egypt, Guatemala, Hungary, India, Indonesia, Jamaica, Malaysia, Mexico, Philippines, Poland, Romania, Serbia, South Africa, Sri Lanka, Thailand, Türkiye, Ukraine.
Countries that have not conducted APPs during 2008-2024: Albania, Armenia, Brazil, Czech Republic, Georgia, Kazakhstan, Norway, Pakistan, Paraguay, Peru, Russia, Singapore, Uruguay.

Table 12: List of Countries under Each Group in Data Collection

Group 1: APP data from central bank webpages and balance sheets (6 economies) Euro Area, United Kingdom, Australia, New Zealand, Chile, Hungary.
Group 2: Central bank balance sheet security holdings data (24 economies) United States, Sweden, Canada, Switzerland, Japan, Türkiye, South Africa, Colombia, Israel, India, Korea, Malaysia, Philippines, Thailand, Poland, Romania, Iceland, Costa Rica, Guatemala, Sri Lanka, Jamaica, Egypt, Argentina, Ukraine.
Group 3: Central bank balance sheet claims on the government and other sectors (3 economies) China, Croatia, Serbia.
Group 4: Other sources (2 economies) Indonesia, Mexico.

Table 13: Variable Description and Data Sources

Variable	Description	Data source
Asset purchase program timeline	Indicator of asset purchase program activity: The variable records net buying, neither buying nor unwinding, and unwinding phases (buying = 1, neither = 0, unwind = −1)	Central banks and authors' calculations
Asset purchase program amount	Net purchase amount and total asset purchase program holdings.	Central banks and other official sources; authors' calculations
Asset purchase program duration	Number of months an asset purchase program buying phase lasts	Central banks and authors' calculations
Advanced economy dummy	A dummy variable for advanced economies	IMF World Economic Outlook
Policy rate	Central bank monetary policy interest rate, in percent per annum	BIS; IMF International Financial Statistics
CPI	Consumer price index measuring headline consumer price inflation	IMF World Economic Outlook
Financial development index	Composite index of the depth, access, and efficiency of financial institutions and markets, scaled between zero and one	IMF Financial Development Index database
General government debt to GDP	General government debt in percent of GDP	IMF World Economic Outlook
Real GDP growth	Year-over-year percent change in real gross domestic product	IMF World Economic Outlook
Sovereign bond yield volatility	Monthly volatility of daily 10-year sovereign local-currency bond yield	Bloomberg
Unemployment rate	Number of unemployed persons in percent of the total labor force	IMF World Economic Outlook
Zero lower bound dummy	Equals one when the central bank policy rate is lower than 0.25 percent	IMF International Financial Statistics; authors' calculations

Supplementary Note on the Central Bank Asset Purchase Database – Data Notes by Country

The paper collects APP data following a decision-tree approach, which classifies countries by progressively assessing data availability for central bank asset purchases. We first determine whether the central bank publishes clean, program-level APP data. If not, we examine whether APP can be proxied using securities holdings reported on the balance sheet. When securities holdings are unavailable, we assess whether claims on the government provide a usable proxy. In the absence of that, we rely on alternative external data sources, such as mirror data from the Ministry of Finance and Debt Management Office annual report data.

To isolate the APP effect, the paper constructs adjusted series by level-adjusting selected countries' securities holdings: a constant equal to the pre-APP stock is subtracted, thereby normalizing the level to zero at the start of the program while preserving all subsequent dynamics. These countries include Canada, China, Colombia, Guatemala, India, Jamaica, Korea, Sri Lanka, Malaysia, Thailand, Türkiye, the Philippines, Ukraine, and South Africa.

United States (USA)

The U.S. Federal Reserve implemented multiple APPs in response to the Global Financial Crisis and the COVID-19 pandemic. APP1 (September 2008–March 2010) involved purchases of government securities, agency bonds, and covered bonds. APP2 (November 2010–June 2011) focused on government securities. Operation Twist (October 2011–December 2012) entailed selling short-term government securities and buying long-term government securities without expanding the balance sheet. APP3 (September 2012–October 2014) combined purchases of government securities and agency bonds. APP4 (March 2020–April 2022) included purchases of government securities and agency bonds. Corporate bonds were purchased via secondary-market purchases of corporate bond ETFs and individual bonds through the Primary and Secondary Market Corporate Credit Facilities (PMCCF and SMCCF). The Federal Reserve undertook passive quantitative tightening from 2022.

Data are retrieved from the securities-holdings data in the Federal Reserve balance sheet at face value, with a weekly frequency, in millions of USD. The paper aggregates the data to monthly frequency using the last available observation in each month. Because the Federal Reserve reports purchases of corporate bonds and corporate bond ETFs together, the two items are merged into one column under "corporate bonds". Security-level holdings data are available from the Federal Reserve Bank of New York: https://www.newyorkfed.org/markets/soma-holdings

Euro Area (EA)

The European Central Bank (ECB) implemented several APPs in response to successive crises, notably the Global Financial Crisis, the euro area sovereign debt crisis, and the COVID-19 pandemic. Earlier APPs included the first two rounds of the Covered Bond Purchase Programme (CBPP1, July 2009–June 2010; CBPP2, November 2011–October 2012), which focused solely on bank-issued covered bonds, and the Securities Markets Programme (SMP, May 2010–September 2012), which aimed to stabilize sovereign debt markets by purchasing stressed euro-area government bonds. Later, the ECB launched the Asset Purchase Programme (APP, October 2014–June 2022), comprising the Public Sector Purchase Programme (PSPP), Covered Bond Purchase Programme 3 (CBPP3), Asset-Backed Securities Purchase Programme (ABSPP), and Corporate Sector Purchase Programme (CSPP). These sub-programmes targeted euro area sovereign bonds, European Union supranational bonds, covered bonds, asset-backed securities, and investment-grade corporate bonds and commercial papers, respectively. The Pandemic Emergency Purchase Programme (PEPP, March 2020–March 2022) was launched in response to the COVID-19 shock, allowing flexible purchases across asset classes and jurisdictions.

Data are reported at the program level on the ECB's website at amortized cost, with a monthly or lower frequency, in million EUR. The ECB's daily liquidity management data and historical weekly financial statements complement the SMP, CBPP1, and CBPP2 data. Because the ECB publishes holdings by program rather than by asset class, central government, sub-sovereign, and agency securities are grouped under "central government securities", while non-financial corporate bonds and commercial paper are grouped under "non-financial corporate bonds".

United Kingdom (GBR)

The Bank of England conducted three major APP rounds in response to systemic shocks: the Global Financial Crisis, the Brexit referendum, and the COVID-19 pandemic. APP1 (March 2009–January 2010) primarily involved purchases of UK government bonds, commercial paper, and a small portfolio of non-financial corporate bonds to restore market functioning. APP2 (October 2016–July 2017) followed the Brexit referendum and reintroduced corporate bond purchases through the Corporate Bond Purchase Scheme (CBPS), alongside further gilt purchases. APP3 (March 2020–October 2022), in response to the COVID-19 crisis, included a sharp expansion in gilt and corporate bond purchases to support monetary policy and market liquidity. In a separate, temporary intervention, the Bank of England purchased long-dated UK government bonds (September to October 2022) to restore orderly market conditions in the long-dated gilt market.

Data are reported at the program level on the Bank of England website using face-value accounting, with a weekly frequency, in millions of GBP. The paper aggregates the data to monthly frequency using the last available observation in each month. Data are collected from the Bank of

England's weekly Asset Purchase Facility (APF) data and APF Quarterly Reports. Security-level gilt holdings data are available from the Bank of England: https://www.bankofengland.co.uk/markets/bank-of-england-market-operations-guide/results-and-usage-data

Sweden (SWE)

The Riksbank's first APP (February 2015–December 2017) purchased Swedish government bonds, including nominal and inflation-linked bonds. In the second APP cycle (March 2020–December 2021), purchases expanded to government bonds, municipal bonds, covered bonds, Treasury bills, corporate bonds, and commercial paper.

Securities-holdings data from the Riksbank balance sheet are recorded at face value on the transaction date, with a monthly frequency, in millions of SEK. The Riksbank prepares its official financial statements using market valuation principles, which differ from the collected monitoring statistics.

Canada (CAN)

The Bank of Canada implemented an APP (March 2020–October 2021) in response to the COVID-19 crisis. Purchases included Government of Canada bonds, provincial bonds, Canada Mortgage Bonds, investment-grade corporate bonds, commercial paper, and bankers' acceptances.

The Bank of Canada publishes balance sheet securities-holdings statistics at a monthly frequency, in million CAD, reflecting holdings based on the Bank's intended use and classification of assets. Security-level holdings data are available from the Bank of Canada: https://www.bankofcanada.ca/markets/government-securities-auctions/bank-of-canada-holdings/historical-bank-of-canada-ho

Switzerland (CHE)

The Swiss National Bank (SNB) undertook targeted asset purchases (March 2009–September 2009) during the Global Financial Crisis, purchasing Swiss-franc covered bonds and non-bank corporate bonds. During the euro area crisis, it repurchased SNB Bills to inject Swiss franc liquidity (August 2011–July 2012). While not a traditional APP, the operation similarly eased domestic financial conditions.

CHF securities-holdings data are sourced from the SNB balance sheet under assets at a monthly frequency, in million Swiss francs, while buyback data come from debt securities holdings under liabilities. Because the SNB does not further disaggregate CHF holdings, covered bonds and non-bank corporate bonds are aggregated under "covered bonds".

Japan (JPN)

The Bank of Japan (BoJ) implemented multiple APPs. The Quantitative Easing Policy, which focused on Japanese Government Bonds (JGBs), was introduced in March 2001 and terminated in March 2006. Purchases of Asset-Backed Securities were initiated in June 2003. The Comprehensive Monetary Easing (October 2010–April 2013) expanded purchases to JGBs, commercial paper, corporate bonds, exchange-traded funds, and Japanese real estate investment trusts. Under the Quantitative and Qualitative Monetary Easing (April 2013–September 2016), the BoJ increased the scale and duration of JGB purchases and continued buying ETFs and J-REITs. This program was followed by the Quantitative and Qualitative Monetary Easing with Yield Curve Control (September 2016–March 2024). In 2024, the BoJ reduced the purchase amount of JGBs, discontinued purchases of ETFs and J-REITs, and lowered the amount of purchases of commercial papers and corporate bonds.

Securities holdings data are obtained from the monthly Bank of Japan balance sheets, in units of JPY 100 million. Fixed-income securities are recorded at amortized cost; ETFs and REITs are stated at cost determined by the moving-average method and reflect impairment revaluation at the end of March and September. Security-level government bond holdings data are available from the Bank of Japan's website: https://www.boj.or.jp/en/statistics/boj/other/mei/index.htm

Türkiye (TUR)

The Central Bank of the Republic of Türkiye launched its first APP (March 2020–July 2020), purchasing government securities in the secondary market. The Government Bond Purchase Program (April 2022–May 2023), introduced under the "Liraization Strategy", used outright auctions to buy government bonds and Treasury lease certificates. Purchases were suspended around May 2023. Securities-holdings data are published from the CBRT balance sheet at face value, monthly, in million TRY.

Australia (AUS)

The Reserve Bank of Australia launched two APPs in response to COVID-19. Long-dated Open Market Operations (March 2020–November 2021) purchased Australian Government Securities and state/territory bonds; the Bond Purchase Program (November 2020–February 2022) purchased longer-dated Australian Government Securities and state-issued semi-government bonds. The RBA ended its APP in February 2022 by allowing holdings to mature without reinvestment.

Program-level data are published at face value, in million AUD. The paper aggregates security face values at each month-end. Security-level data are available from the Reserve Bank of Australia's

website: https://www.rba.gov.au/statistics/tables/

New Zealand (NZL)

The Reserve Bank of New Zealand implemented the Large-Scale Asset Purchase (LSAP) program during March 2020–July 2021, purchasing nominal and inflation-indexed government bonds and Local Government Funding Agency bonds. Purchases ended in July 2021; portfolio reduction through maturities and managed sales began in July 2022. The central bank also conducted small-scale purchases of nominal New Zealand government and Local Government Funding Agency bonds under its Bond Market Liquidity Support (BMLS) in March and April 2020, and unwinding began in May 2021.

Securities are accounted for on a fair value basis with the interest component of the return accounted for using the effective interest rate method. Data are monthly in millions of NZD. The published LSAP data combine central and local government bonds in the balance sheet, while the BMLS data provide separate central and local government bond time series. Therefore, in the dataset, the "central government bonds" item includes both central and local government bonds from the LSAP program and the central government bonds from the BMLS program, and the "local government bonds" item only includes local government bonds from the BMLS program.

South Africa (ZAF)

The South African Reserve Bank implemented the Bond Purchase Program (March 2020–January 2021), purchasing central government bonds across the yield curve and subsequently allowing holdings to mature. Investments in central government securities are published from the SARB balance sheet at fair value in millions of ZAR.

Chile (CHL)

The Central Bank of Chile implemented a targeted APP (March 2020–May 2021), purchasing bank bonds and central bank bonds. It began with the Cash Purchase/ Forward Sale Program and expanded into outright bank-bond purchases; it also repurchased BCCh-issued bonds in mid-2020.

Buyback amounts in millions of USD are converted to Chilean pesos using the monthly BCCh exchange rate. Monthly claims on depository institutions include bank bonds and certificates of deposit; the statistics are merged under "bank bonds".

Colombia (COL)

The Banco de la República has not conducted a conventional APP, but undertook limited purchases (March 2020–June 2020) of government bonds and high-quality private-sector debt issued by financial institutions to stabilize market liquidity. Securities holdings are published on the balance sheet at amortized cost, in billion COP.

Mexico (MEX)

The Bank of Mexico has not conducted a conventional APP. Operation Twist (April 2020–June 2020) purchased long-term and sold short-term government securities without a net expansion of holdings. Data are collected from BIS Papers No. 122 (2021), "Unconventional Monetary Policy Tools in Emerging Market Economies".

Israel (ISR)

The Bank of Israel launched a domestic APP (March 2020–December 2021), purchasing government bonds and later investment-grade corporate bonds to improve liquidity and credit conditions. The balance sheet reports domestic securities together at amortized cost, in million ILS. Government and non-financial corporate bonds are merged under "central government bonds".

India (IND)

The Reserve Bank of India launched a domestic APP (March 2020–November 2021) involving long-term government-security purchases through open market operations and the Government Securities Acquisition Program. Operation Twist (December 2019–around May 2021) purchased long-term and sold short-term government securities. Information on the Reserve Bank of India (RBI)'s holdings of government debt securities is published monthly in the DBIE website (Statistics -> Financial Sector -> Monetary Statistics ->Reserve Bank of India Survey) in rupee crores at fair value.

Indonesia (IDN)

Bank Indonesia launched a domestic APP (July 2020–December 2022). Primary-market purchases of central government securities were agreed with the Ministry of Finance under burden-sharing arrangements from 2020 to 2022, and partial unwinding followed through end-2024. From August 2022 to September 2023, BI also conducted operational twist to sell short-term securities while

purchasing long-term securities.

Government bond ownership data are published by the Ministry of Finance at face value, in trillion rupiah. Before 2016, they cover total BI holdings; from January 2016 they exclude market-operation holdings. Indonesia's pandemic-era Treasury securities are non-marketable, and the Ministry of Finance ownership data do not distinguish between marketable and non-marketable securities. While Bank Indonesia publicly announced the use of Operation Twist, the central bank does not publish any statistics on the size of the operation.

Korea (KOR)

The Bank of Korea purchased government bonds during the Global Financial Crisis (November 2008) and conducted outright Treasury bond purchases during COVID-19 (March 2020–June 2021). From July 2020, it lent to the Corporate Liquidity Support Organization, a Special Purpose Vehicle (SPV) purchasing corporate bonds and commercial paper.

Government-bond investments are published at amortized cost, monthly, in billions of KRW. We use the Bank of Korea's main accounts data and take the series "Loans and Discounts - Other Loans" from June 2020 to February 2024 to proxy the SPV's corporate-bond and commercial-paper holdings.

Malaysia (MYS)

Bank Negara Malaysia purchased RM4.3 billion in government bonds (November 2016–August 2017). During COVID-19, it conducted an APP (March–December 2020) involving Malaysian Government Securities and Government Investment Issues, with purchases limited to 10 percent of an individual security's outstanding volume. Government-paper holdings are published monthly on BNM's balance sheet in millions of MYR, at face value and amortized cost.

Philippines (PHL)

The Bangko Sentral ng Pilipinas implemented an APP (March 2020–December 2021) involving secondary-market government-security purchases. A PHP 300 billion repurchase agreement with the Treasury is excluded. Government-security holdings are released monthly, in million PHP, at amortized cost.

Thailand (THA)

The Bank of Thailand implemented an APP (March–April 2020) via government-security purchases and temporary buybacks of BoT bonds. The Bond Stabilization Fund (April 2020–December 2022), intended to purchase investment-grade corporate paper, was never utilized. Monthly BoT-bond and government-bond holdings are published in millions of THB at amortized cost. The March–April 2020 increase in BoT-bond holdings proxies buybacks.

China (CHN)

The People's Bank of China (PBoC) bought short-term and sold longer-term central government bonds during August-December 2024. The PBoC's government bond trading operations were positioned as a tool for base money injection and liquidity management. These operations were conducted in conjunction with other monetary policy instruments to help maintain an appropriate monetary and financial environment. The primary purpose of the operations is monetary policy implementation, while also helping to maintain orderly market functioning. Government-bond holdings are published monthly in billion CNY under "Claims on Central Government" at amortized cost. Our APP series based on PBoC's balance sheet data differs slightly from the announced amounts by the PBoC.

Serbia (SRB)

The National Bank of Serbia purchased dinar-denominated government securities in the secondary market (April–September 2020). In May 2020 it temporarily made certain domestic corporate bonds eligible for repo and outright purchases. Monthly local-currency claims on government are collected from the balance sheet in millions of RSD at amortized cost.

Hungary (HUN)

The Magyar Nemzeti Bank implemented the Mortgage Bond Purchase Programme I (January–December 2018), Bond Funding for Growth Scheme (September 2019–June 2022), Mortgage Bond Purchase Programme II (May 2020–July 2021), Government Securities Purchase Programme (May 2020–December 2021), and Green Mortgage Bond Purchase Programme (February 2021–March 2022). Mortgage bonds were purchased in primary and secondary markets.

Program-level data are published at amortized cost, in billions of HUF, combining month-end security face values and monthly balance-sheet items. Security-level data are available from MNB: https://www.mnb.hu/en/monetary-policy/monetary-policy-instruments

Croatia (HRV)

The Croatian National Bank conducted outright secondary-market purchases of central government bonds (March–July 2020) through five auctions to support liquidity and public financing conditions. Holdings are classified as "claims on general government" and measured monthly at amortized cost.

Poland (POL)

The Narodowy Bank Polski introduced a domestic APP (April 2020–November 2021), purchasing government securities through Structural Open Market Operations and later adding state-backed agency bonds. Monthly government-bond holdings are collected from the balance sheet in millions of PLN at amortized cost.

Romania (ROU)

The National Bank of Romania purchased central government securities (March–August 2020) through bilateral operations and conducted ad hoc purchases in March–May 2022 and March 2023. Monthly government-bond holdings are collected from the balance sheet in millions of RON at amortized cost.

Iceland (ISL)

The Central Bank of Iceland acquired government and Housing Financing Fund securities during episodes in 2011, 2016, and 2017 linked to capital-account liberalization. During COVID-19 it launched a formal Treasury-bond purchase program (April 2020–August 2021).

Monthly government-bond holdings are collected from the balance sheet in millions of ISK at amortized cost. Central bank certificates and HFF bonds are not separately reported. Security-level data are available at https://databank.is/report/inactivemeasures?from=2016-06-11&to=2026-06-11&page=MEASURES.TREASURYBONDS.TABLE

Costa Rica (CRI)

The Central Bank of Costa Rica purchased central government bonds in the secondary market (June 2020–August 2021) as part of extraordinary liquidity measures to mitigate financial tensions. Domestic-security investments are published monthly at amortized cost.

Egypt (EGY)

The Central Bank of Egypt announced a stock purchase program in March 2020, committing up to EGP 20 billion to support the domestic exchange market. Monthly statistics are not published; instead, biannual statistics are disclosed in the central bank's annual financial statements report under "Financial Assets – Local Portfolio Managed by Third Parties".

Guatemala (GTM)

The Banco de Guatemala purchased about GTQ 100 million in Treasury bonds in November-December 2008 and reactivated secondary-market purchases during April 2020–March 2021. Holdings are reported monthly in GTQ at amortized cost under balance sheet item "Inversiones en valores internos".

Sri Lanka (LKA)

The Central Bank of Sri Lanka purchased government bonds in primary and secondary markets (March 2020–September 2023) to finance the government, support activity, and restrain domestic yields. Government securities holdings are retrieved from the central bank balance sheet, reported in billion LKR at face value.

Jamaica (JAM)

The Bank of Jamaica purchased GOJ securities in the secondary market (March–October 2020) to support liquidity. Balance sheet holdings are published biweekly in thousands of JMD at amortized cost.

Argentina (ARG)

The Central Bank of Argentina (BCRA) reduced its holdings of government securities from 2018 under the IMF's Stand-By Arrangement and then resumed monetary financing from December 2019 amid a deterioration in public finances. In addition to direct monetization of the fiscal deficit through temporary advances and profit transfers, the BCRA enabled indirect monetization in 2022 and 2023 by purchasing Treasury bonds in the secondary market to support their prices and establishing a liquidity window to guarantee banks the possibility of selling their portfolios of government securities to the central bank. In December 2023, a stabilization plan was implemented by the new administration with the stated objective of ending monetary financing by the BCRA

and reducing the BCRA's exposure to public sector assets. The BCRA publishes weekly balance sheet information on its holdings of domestic securities denominated in thousands of Argentine pesos and measured at amortized cost.

Ukraine (UKR)

Following Russia's war with Ukraine, the National Bank of Ukraine purchased domestic government "war bonds" directly in the primary market (February 2022–March 2023) to finance emergency budget needs. As multilateral financing increased, it ceased net purchases and committed to avoiding further monetary financing. Central government securities are reported monthly at amortized cost, in millions of UAH, including domestic government bonds and crisis-related recapitalization or special-purpose bonds.



PUBLICATIONS

How Do Monetary Authorities Buy Publicly Traded Financial Assets? A Cross-Country Database and an Empirical Study of Central Bank Asset Purchase Programs

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